## THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)  (part 1 of 4)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w111_c | 20 | White | | Injector 4 drive (-) | DRB102 receptacle (RH firewall) pin 24 | 630 | ST185-AB-cabin | INJ4 |
| **2** | w110_c | 20 | White | | Injector 3 drive (-) | DRB102 receptacle (RH firewall) pin 23 | 630 | ST185-AB-cabin | INJ3 |
| **3** | w109_c | 20 | White | | Injector 2 drive (-) | DRB102 receptacle (RH firewall) pin 22 | 630 | ST185-AB-cabin | INJ2 |
| **4** | w108_c | 20 | White | | Injector 1 drive (-) | DRB102 receptacle (RH firewall) pin 21 | 630 | ST185-AB-cabin | INJ1 |
| **8** | w200_c | 20 | White | | Crank sensor signal | DRB102 receptacle (RH firewall) pin 74 | 630 | ST185-AB-cabin | CRANKC |
| **10** | w123_c | 20 | White | | Coil 4 trigger (IGT) | DRB102 receptacle (RH firewall) pin 12 | 630 | ST185-AB-cabin | COP4 |
| **11** | w122_c | 20 | White | | Coil 3 trigger (IGT) | DRB102 receptacle (RH firewall) pin 11 | 630 | ST185-AB-cabin | COP3 |
| **12** | w121_c | 20 | White | | Coil 2 trigger (IGT) | DRB102 receptacle (RH firewall) pin 10 | 630 | ST185-AB-cabin | COP2 |
| **13** | w120_c | 20 | White | | Coil 1 trigger (IGT) | DRB102 receptacle (RH firewall) pin 9 | 630 | ST185-AB-cabin | COP1 |
| **15** | w201_c | 20 | White | | Coolant temp signal | DRB102 receptacle (RH firewall) pin 70 | 630 | ST185-AB-cabin | ECT |
| **16** | w202_c | 20 | White | | Intake temp signal | DRB102 receptacle (RH firewall) pin 33 | 630 | ST185-AB-cabin | IATC |
| **17** | w26_c | 20 | White | | MAP sensor signal | DRB102 receptacle (RH firewall) pin 17 | 630 | ST185-AB-cabin | MAPC |
| **21** | w103_c | 20 | White | | Boost solenoid (-) | DRB102 receptacle (RH firewall) pin 58 | 630 | ST185-AB-cabin | BOOST |
| **22** | w128_c | 20 | White | | Throttle pos 1 main | DRB102 receptacle (RH firewall) pin 42 | 630 | ST185-AB-cabin | ETBS |
| **27** | w_mrs_c | 20 | White | | EHPS pump speed | DRB102 receptacle (RH firewall) pin 84 | 630 | ST185-AB-cabin | MRSSPD |
| **30** | w16_c | 20 | White | | Turbo speed signal | DRB102 receptacle (RH firewall) pin 53 | 630 | ST185-AB-cabin | TURBO |
| **31** | w48_c | 20 | White | | Flex fuel signal | DRB102 receptacle (RH firewall) pin 31 | 630 | ST185-AB-cabin | FLEX |
| **33** | w129_c | 20 | White | | Throttle pos 2 sub | DRB102 receptacle (RH firewall) pin 44 | 630 | ST185-AB-cabin | ETBS |

### ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **6** | w203_c | 20 | White | | Oil temp signal | DRB102 receptacle (RH firewall) pin 61 | 630 | ST185-AB-cabin | OILT |
| **7** | w204_c | 20 | White | | Charge temp signal | DRB102 receptacle (RH firewall) pin 59 | 630 | ST185-AB-cabin | CIAT |
| **9** | w205_c | 20 | White | | Knock sensor signal | DRB102 receptacle (RH firewall) pin 71 | 630 | ST185-AB-cabin | KNOCK |
| **15** | w29_c | 20 | White | | Oil pressure signal | DRB102 receptacle (RH firewall) pin 28 | 630 | ST185-AB-cabin | OILPC |
| **16** | w32_c | 20 | White | | Fuel pressure signal | DRB102 receptacle (RH firewall) pin 14 | 630 | ST185-AB-cabin | FUELP |
| **18** | w124_c | 18 | White | | ETB motor + | DRB102 receptacle (RH firewall) pin 45 | 630 | ST185-AB-cabin | ETBM |
| **23** | w35_c | 20 | White | | Coolant press signal | DRB102 receptacle (RH firewall) pin 26 | 630 | ST185-AB-cabin | CLNTP |
| **26** | w215_c | 18 | White | | ETB motor - | DRB102 receptacle (RH firewall) pin 46 | 630 | ST185-AB-cabin | ETBM |
| **29** | w_vss_sig_c | 20 | White | | Gearbox speed signal | DRB102 receptacle (RH firewall) pin 68 | 630 | ST185-AB-cabin | VSS |

### BUS-5V - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_rail_sp_5v_c | 20 | Orange | | +5 V throttle sensor | DRB102 receptacle (RH firewall) pin 41 | 520 | ST185-AB-cabin | 5V |
| **3** | w_5v_a_c39 | 20 | Orange | | +5 V MAP sensor | DRB102 receptacle (RH firewall) pin 16 | 520 | ST185-AB-cabin | 5V |
| **4** | w_5v_a_c40 | 20 | Orange | | +5 V turbo speed | DRB102 receptacle (RH firewall) pin 52 | 520 | ST185-AB-cabin | 5V |
| **5** | w_5v_brail_c45 | 20 | Orange | | +5 V oil pressure | DRB102 receptacle (RH firewall) pin 27 | 520 | ST185-AB-cabin | 5V |
| **6** | w_5v_brail_c46 | 20 | Orange | | +5 V fuel pressure | DRB102 receptacle (RH firewall) pin 13 | 520 | ST185-AB-cabin | 5V |
| **7** | w_5v_brail_c47 | 20 | Orange | | +5 V coolant press | DRB102 receptacle (RH firewall) pin 25 | 520 | ST185-AB-cabin | 5V |

### BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_rail_sp_gndout_c | 20 | Blue | | Gnd IAT/TPS/coolant | DRB102 receptacle (RH firewall) pin 34 | 520 | ST185-AB-cabin | SGND |
| **3** | w_gnd_a_c41 | 20 | Blue | | Gnd crank sensor | DRB102 receptacle (RH firewall) pin 75 | 520 | ST185-AB-cabin | SGND |
| **4** | w_gnd_a_c42 | 20 | Blue | | Gnd cam sensor | DRB102 receptacle (RH firewall) pin 65 | 520 | ST185-AB-cabin | SGND |
| **5** | w_gnd_a_c43 | 20 | Blue | | Gnd turbo speed | DRB102 receptacle (RH firewall) pin 56 | 520 | ST185-AB-cabin | SGND |
| **6** | w_gnd_a_c44 | 20 | Blue | | Gnd flex + MAP | DRB102 receptacle (RH firewall) pin 18 | 520 | ST185-AB-cabin | SGND |

### BUS-GNDOUT-B - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **6** | w12_c | 20 | Blue | | Knock sensor (-) | DRB102 receptacle (RH firewall) pin 72 | 520 | ST185-AB-cabin | KNOCK |
| **7** | w_gnd_b_c19 | 20 | Blue | | Gnd oil/clnt/temps | DRB102 receptacle (RH firewall) pin 29 | 520 | ST185-AB-cabin | SGND |
| **8** | w_gnd_b_c21 | 20 | Blue | | Gnd fuel press + VSS | DRB102 receptacle (RH firewall) pin 15 | 520 | ST185-AB-cabin | SGND |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)  (part 1 of 4)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 1 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)  (part 2 of 4)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-8V - ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rail_sp_8v_c | 20 | Orange | | +8 V to cam sensor | DRB102 receptacle (RH firewall) pin 63 | 520 | ST185-AB-cabin | 8V |

### SP-CAM - At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_cam_sig_c | 20 | White | | Cam sensor signal | DRB102 receptacle (RH firewall) pin 64 | 520 | ST185-AB-cabin | CAM |

### SP-CANH - CAN H join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_h_bh_bc | 20 | White | | CAN H to CAN-Lambda | DRB102 receptacle (RH firewall) pin 79 | 680 | ST185-AB-cabin | CAN |

### SP-CANL - CAN L join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_l_bh_bc | 20 | Green | | CAN L to CAN-Lambda | DRB102 receptacle (RH firewall) pin 80 | 680 | ST185-AB-cabin | CAN |

### SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rev_sw_c | 16 | Yellow | | Reverse sw +12 V out | DRB102 receptacle (RH firewall) pin 78 | 520 | ST185-AB-cabin | REV |

### SP-SW12 - ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rail_sp_sw12_c | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | DRB102 receptacle (RH firewall) pin 30 | 520 | ST185-AB-cabin | SW12 |
| | w_rail_sw12_b_c_ac | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | DRB102 receptacle (RH firewall) pin 67 | 520 | ST185-AB-cabin | SW12 |

**REVISIONS**

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**ST185-harness - every wiring harness of the car, one drawing set**

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)  (part 2 of 4)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 2 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)  (part 3 of 4)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018  + DRBF-1A  |  contacts 0460-220-1231  |  pins per TE drawing DRB12-102PXE-L018 rev C3

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_cop_gnd_ac | 14 | Black | | Coil ground to the POWER box NEG bar | POWER box NEG bar M8 stud (glove box) | 450 | ST185-AB-cabin | COPG |
| **9** | w120_c | 20 | White | | Coil 1 trigger (IGT) | Link G4X plug A (centre dash) pin 13 | 630 | ST185-AB-cabin | COP1 |
| **10** | w121_c | 20 | White | | Coil 2 trigger (IGT) | Link G4X plug A (centre dash) pin 12 | 630 | ST185-AB-cabin | COP2 |
| **11** | w122_c | 20 | White | | Coil 3 trigger (IGT) | Link G4X plug A (centre dash) pin 11 | 630 | ST185-AB-cabin | COP3 |
| **12** | w123_c | 20 | White | | Coil 4 trigger (IGT) | Link G4X plug A (centre dash) pin 10 | 630 | ST185-AB-cabin | COP4 |
| **13** | w_5v_brail_c46 | 20 | Orange | | +5 V fuel pressure | DT06-12 plug (ECU splice point) pin 6 | 520 | ST185-AB-cabin | 5V |
| **14** | w32_c | 20 | White | | Fuel pressure signal | ECU-B (Link G4X plug B, centre dash) pin 16 | 630 | ST185-AB-cabin | FUELP |
| **15** | w_gnd_b_c21 | 20 | Blue | | Gnd fuel press + VSS | DT06-12 plug (ECU splice point) pin 8 | 520 | ST185-AB-cabin | SGND |
| **16** | w_5v_a_c39 | 20 | Orange | | +5 V MAP sensor | DT06-12 plug (ECU splice point) pin 3 | 520 | ST185-AB-cabin | 5V |
| **17** | w26_c | 20 | White | | MAP sensor signal | Link G4X plug A (centre dash) pin 17 | 630 | ST185-AB-cabin | MAPC |
| **18** | w_gnd_a_c44 | 20 | Blue | | Gnd flex + MAP | DT06-12 plug (ECU splice point) pin 6 | 520 | ST185-AB-cabin | SGND |
| **21** | w108_c | 20 | White | | Injector 1 drive (-) | Link G4X plug A (centre dash) pin 4 | 630 | ST185-AB-cabin | INJ1 |
| **22** | w109_c | 20 | White | | Injector 2 drive (-) | Link G4X plug A (centre dash) pin 3 | 630 | ST185-AB-cabin | INJ2 |
| **23** | w110_c | 20 | White | | Injector 3 drive (-) | Link G4X plug A (centre dash) pin 2 | 630 | ST185-AB-cabin | INJ3 |
| **24** | w111_c | 20 | White | | Injector 4 drive (-) | Link G4X plug A (centre dash) pin 1 | 630 | ST185-AB-cabin | INJ4 |
| **25** | w_5v_brail_c47 | 20 | Orange | | +5 V coolant press | DT06-12 plug (ECU splice point) pin 7 | 520 | ST185-AB-cabin | 5V |
| **26** | w35_c | 20 | White | | Coolant press signal | ECU-B (Link G4X plug B, centre dash) pin 23 | 630 | ST185-AB-cabin | CLNTP |
| **27** | w_5v_brail_c45 | 20 | Orange | | +5 V oil pressure | DT06-12 plug (ECU splice point) pin 5 | 520 | ST185-AB-cabin | 5V |
| **28** | w29_c | 20 | White | | Oil pressure signal | ECU-B (Link G4X plug B, centre dash) pin 15 | 630 | ST185-AB-cabin | OILPC |
| **29** | w_gnd_b_c19 | 20 | Blue | | Gnd oil/clnt/temps | DT06-12 plug (ECU splice point) pin 7 | 520 | ST185-AB-cabin | SGND |
| **30** | w_rail_sp_sw12_c | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | ECU splice point - SP-SW12: switched +12 V (LOGIC SW12) | 520 | ST185-AB-cabin | SW12 |
| **31** | w48_c | 20 | White | | Flex fuel signal | Link G4X plug A (centre dash) pin 31 | 630 | ST185-AB-cabin | FLEX |
| **33** | w202_c | 20 | White | | Intake temp signal | Link G4X plug A (centre dash) pin 16 | 630 | ST185-AB-cabin | IATC |
| **34** | w_rail_sp_gndout_c | 20 | Blue | | Gnd IAT/TPS/coolant | DT06-12 plug (ECU splice point) pin 2 | 520 | ST185-AB-cabin | SGND |
| **41** | w_rail_sp_5v_c | 20 | Orange | | +5 V throttle sensor | DT06-12 plug (ECU splice point) pin 2 | 520 | ST185-AB-cabin | 5V |
| **42** | w128_c | 20 | White | | Throttle pos 1 main | Link G4X plug A (centre dash) pin 22 | 630 | ST185-AB-cabin | ETBS |
| **44** | w129_c | 20 | White | | Throttle pos 2 sub | Link G4X plug A (centre dash) pin 33 | 630 | ST185-AB-cabin | ETBS |
| **45** | w124_c | 18 | White | | ETB motor + | ECU-B (Link G4X plug B, centre dash) pin 18 | 630 | ST185-AB-cabin | ETBM |
| **46** | w215_c | 18 | White | | ETB motor - | ECU-B (Link G4X plug B, centre dash) pin 26 | 630 | ST185-AB-cabin | ETBM |
| **52** | w_5v_a_c40 | 20 | Orange | | +5 V turbo speed | DT06-12 plug (ECU splice point) pin 4 | 520 | ST185-AB-cabin | 5V |
| **53** | w16_c | 20 | White | | Turbo speed signal | Link G4X plug A (centre dash) pin 30 | 630 | ST185-AB-cabin | TURBO |
| **56** | w_gnd_a_c43 | 20 | Blue | | Gnd turbo speed | DT06-12 plug (ECU splice point) pin 5 | 520 | ST185-AB-cabin | SGND |
| **58** | w103_c | 20 | White | | Boost solenoid (-) | Link G4X plug A (centre dash) pin 21 | 630 | ST185-AB-cabin | BOOST |
| **59** | w204_c | 20 | White | | Charge temp signal | ECU-B (Link G4X plug B, centre dash) pin 7 | 630 | ST185-AB-cabin | CIAT |
| **61** | w203_c | 20 | White | | Oil temp signal | ECU-B (Link G4X plug B, centre dash) pin 6 | 630 | ST185-AB-cabin | OILT |
| **63** | w_rail_sp_8v_c | 20 | Orange | | +8 V to cam sensor | ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6) | 520 | ST185-AB-cabin | 8V |
| **64** | w_cam_sig_c | 20 | White | | Cam sensor signal | At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V | 520 | ST185-AB-cabin | CAM |
| **65** | w_gnd_a_c42 | 20 | Blue | | Gnd cam sensor | DT06-12 plug (ECU splice point) pin 4 | 520 | ST185-AB-cabin | SGND |
| **67** | w_rail_sw12_b_c_ac | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | ECU splice point - SP-SW12: switched +12 V (LOGIC SW12) | 520 | ST185-AB-cabin | SW12 |
| **68** | w_vss_sig_c | 20 | White | | Gearbox speed signal | ECU-B (Link G4X plug B, centre dash) pin 29 | 630 | ST185-AB-cabin | VSS |
| **70** | w201_c | 20 | White | | Coolant temp signal | Link G4X plug A (centre dash) pin 15 | 630 | ST185-AB-cabin | ECT |
| **71** | w205_c | 20 | White | | Knock sensor signal | ECU-B (Link G4X plug B, centre dash) pin 9 | 630 | ST185-AB-cabin | KNOCK |
| **72** | w12_c | 20 | Blue | | Knock sensor (-) | DT06-12 plug (ECU splice point) pin 6 | 520 | ST185-AB-cabin | KNOCK |
| **74** | w200_c | 20 | White | | Crank sensor signal | Link G4X plug A (centre dash) pin 8 | 630 | ST185-AB-cabin | CRANKC |
| **75** | w_gnd_a_c41 | 20 | Blue | | Gnd crank sensor | DT06-12 plug (ECU splice point) pin 3 | 520 | ST185-AB-cabin | SGND |
| **77** | w_rev_feed_c | 16 | Red | | Back-up sw +12 V in | OEM IG2 pin 4 (under the ECU, cowl side) - factory back-up SW feed pin 4 | 630 | ST185-AB-cabin | REV |
| **78** | w_rev_sw_c | 16 | Yellow | | Reverse sw +12 V out | SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps | 520 | ST185-AB-cabin | REV |
| **79** | wc_h_bh_bc | 20 | White | | CAN H to CAN-Lambda | CAN H join - ECU, CSB3, cluster, Pi, bulkhead | 680 | ST185-AB-cabin | CAN |
| **80** | wc_l_bh_bc | 20 | Green | | CAN L to CAN-Lambda | CAN L join - ECU, CSB3, cluster, Pi, bulkhead | 680 | ST185-AB-cabin | CAN |
| **83** | w_fb_mrs_en | 20 | Red | | EFI relay k_efi -> LOGIC MRS 7.5 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 450 | ST185-AB-cabin | EPSEN |
| **84** | w_mrs_c | 20 | White | | EHPS pump speed | Link G4X plug A (centre dash) pin 27 | 630 | ST185-AB-cabin | MRSSPD |
| **85** | w_eps_rly | 18 | White | | EHPS relay coil (-) | B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro... | 450 | ST185-AB-cabin | EPS |
| **96** | w_lamg_ac | 14 | Black | | CAN-Lambda ground | POWER box NEG bar M8 stud (glove box) | 450 | ST185-AB-cabin | LAM |

### K-EPS - B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to ground by the MR-S pump pin B5 (R22); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole plug (TE terminals 1/2, not marked 85/86), res...

housing V23132-A2001-B200  |  contacts 282110-1  |  pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **85** | w_eps_rly | 18 | White | | EHPS relay coil (-) | DRB102 receptacle (RH firewall) pin 85 | 450 | ST185-AB-cabin | EPS |

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| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 3 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)  (part 4 of 4)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968849-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **BUS2-12** | w_fb_mrs_en | 20 | Red | | EFI relay k_efi -> LOGIC MRS 7.5 A fuse | DRB102 receptacle (RH firewall) pin 83 | 450 | ST185-AB-cabin | EPSEN |

### POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)

terminal 320575

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_lamg_ac | 14 | Black | | CAN-Lambda ground | DRB102 receptacle (RH firewall) pin 96 | 450 | ST185-AB-cabin | LAM |
| | w_cop_gnd_ac | 14 | Black | | Coil ground | DRB102 receptacle (RH firewall) pin 4 | 450 | ST185-AB-cabin | COPG |

### OEM IG2 pin 4 (under the ECU, cowl side) - factory back-up SW feed

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_rev_feed_c | 16 | Red | | Back-up sw +12 V in | DRB102 receptacle (RH firewall) pin 77 | 630 | ST185-AB-cabin | REV |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
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| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds)  (part 4 of 4)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 4 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar)  (part 1 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-3 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | w_a5_pin | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak... | ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode) | 310 | ST185-AB-cabin | SW12 |
| **6** | w19 | 20 | Orange | | +8 V sensor supply | ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6) | 310 | ST185-AB-cabin | 8V |
| **24** | w68 | 18 | Blue | | Sensor ground | DT06-12 plug (ECU splice point) pin 1 | 310 | ST185-AB-cabin | SGND |
| **25** | w70 | 18 | Black | | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |
| **32** | w67 | 18 | Orange | | +5 V sensor supply | DT06-12 plug (ECU splice point) pin 1 | 310 | ST185-AB-cabin | 5V |
| **34** | w_a34 | 18 | Black | | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |

### ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-3 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **22** | w69 | 18 | Blue | | Sensor ground B | DT06-12 plug (ECU splice point) pin 1 | 310 | ST185-AB-cabin | SGND |
| **25** | w_b25_ac | 18 | Black | | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |
| **34** | w_b34_ac | 18 | Black | | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 310 | ST185-AB-cabin | GND |

### BUS-5V - DT06-12 plug (ECU splice point)

housing DT06-12SA  + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w67 | 18 | Orange | | +5 V from ECU A32 | Link G4X plug A (centre dash) pin 32 | 310 | ST185-AB-cabin | 5V |
| **8** | w_aps_vc1 | 20 | Red | | +5 V pedal sensor 1 | accelerator pedal sensor pin 4 | 1200 | ST185-AB-cabin | 5V |
| **9** | w210 | 20 | Red | | +5 V pedal sensor 2 | accelerator pedal sensor pin 1 | 1200 | ST185-AB-cabin | 5V |
| **10** | cab_fspur_5v | 20 | Red | | +5 V front VRC | to front VR conditioner pin 1 | 640 | ST185-AB-cabin | 5V |
| **11** | cab_rspur_5v | 20 | Red | | +5 V rear VRC | IX-WS-R inline to rear VRC (dash above ECU) pin 1 | 640 | ST185-AB-cabin | 5V |
| **12** | w_5v_bus_link | 20 | Orange | | +5 V to BUS-5V-2 | DT06-8 plug (ECU splice point) pin 1 | 200 | ST185-AB-cabin | 5V |

### BUS-5V-2 - DT06-8 plug (ECU splice point)

housing DT06-08SA  + W8S; DT04-08PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_5v_bus_link | 20 | Orange | | +5 V from BUS-5V | DT06-12 plug (ECU splice point) pin 12 | 200 | ST185-AB-cabin | 5V |
| **2** | w_cru_5v_ac | 20 | Orange | | +5 V cruise ladder | resistor r_cruise on CSB3 pin 1 | 680 | ST185-AB-cabin | 5V |
| **4** | w44 | 20 | Orange | | +5 V fuel lvl pullup | 470R fuel level pull-up to +5V pin 2 | 200 | ST185-AB-cabin | FLVL |

### BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA  + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w68 | 18 | Blue | | Sensor gnd from A24 | Link G4X plug A (centre dash) pin 24 | 310 | ST185-AB-cabin | SGND |
| **7** | w40 | 20 | Yellow | | Pedal 1 ground | accelerator pedal sensor pin 5 | 1200 | ST185-AB-cabin | SGND |
| **8** | w_aps_gnd2 | 20 | Yellow | | Pedal 2 ground | accelerator pedal sensor pin 2 | 1200 | ST185-AB-cabin | SGND |
| **9** | cab_fspur_gnd | 20 | Yellow | | Front VRC ground | to front VR conditioner pin 2 | 640 | ST185-AB-cabin | SGND |

### BUS-GNDOUT-B - DT06-12 plug (ECU splice point)

housing DT06-12SA  + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w69 | 18 | Blue | | Sensor gnd from B22 | ECU-B (Link G4X plug B, centre dash) pin 22 | 310 | ST185-AB-cabin | SGND |
| **2** | w45 | 20 | Blue | | Fuel level ground | IX-FUEL inline to fuel sender lead (trunk end) pin 2 | 2820 | ST185-AB-cabin | SGND |
| **3** | cab_rspur_gnd | 20 | Yellow | | Rear VRC ground | IX-WS-R inline to rear VRC (dash above ECU) pin 2 | 640 | ST185-AB-cabin | SGND |

### BUS-SW12 - SW12 bus plug DT06-12 (glove box, by LOGIC box)

housing DT06-12SA  + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_fb_sw12 | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | ECU splice point - SP-SW12: switched +12 V (LOGIC SW12) | 570 | ST185-AB-cabin | SW12 |

### SP-8V - ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_cam_pullup_8v | 20 | Orange | | +8 V to cam pull-up | Cam pull-up 1.8 k to +8 V pin 1 | 200 | ST185-AB-cabin | CAM |
| | w19 | 20 | Orange | | +8 V sensor supply | Link G4X plug A (centre dash) pin 6 | 310 | ST185-AB-cabin | 8V |

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| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
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**ST185-harness - every wiring harness of the car, one drawing set**

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar)  (part 1 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 5 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar)  (part 2 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-A5 - ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode)

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w78 | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak... | ECU splice point - SP-SW12: switched +12 V (LOGIC SW12) | 200 | ST185-AB-cabin | SW12 |
| | w_ig2_diode | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak... | IG2 wake diode (1N5822) pin 2 | 200 | ST185-AB-cabin | SW12 |
| | w_a5_pin | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak... | Link G4X plug A (centre dash) pin 5 | 310 | ST185-AB-cabin | SW12 |

### SP-CSB-GND - CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_csbg_ac | 20 | Black | | CSB3 + opto ground | POWER box NEG bar M8 stud (glove box) | 1050 | ST185-AB-cabin | CSB |

### SP-GND-ECU - ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w70 | 18 | Black | | ECU power ground | Link G4X plug A (centre dash) pin 25 | 310 | ST185-AB-cabin | GND |
| | w_a34 | 18 | Black | | ECU power ground | Link G4X plug A (centre dash) pin 34 | 310 | ST185-AB-cabin | GND |
| | w_b25_ac | 18 | Black | | ECU power ground | ECU-B (Link G4X plug B, centre dash) pin 25 | 310 | ST185-AB-cabin | GND |
| | w_b34_ac | 18 | Black | | ECU power ground | ECU-B (Link G4X plug B, centre dash) pin 34 | 310 | ST185-AB-cabin | GND |
| | w66 | 10 | Black | | ECU power ground | POWER box NEG bar M8 stud (glove box) | 570 | ST185-AB-cabin | GND |

### SP-SW12 - ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fb_sw12 | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 2 | 570 | ST185-AB-cabin | SW12 |
| | w78 | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak... | ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode) | 200 | ST185-AB-cabin | SW12 |

### D-IG2-WAKE - IG2 wake diode (1N5822)

inline diode 1N5822

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_ig2_in | 20 | Yellow | | IG2 wake feed | OEM J/B No.1 (LH kick panel) 1H pin 7, B-O pin 7 | 1210 | ST185-AB-cabin | IG2WAKE |
| **2** | w_ig2_diode | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak... | ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode) | 200 | ST185-AB-cabin | SW12 |

### R-CAM-PULLUP - Cam pull-up 1.8 k to +8 V

inline resistor 1.8k 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_cam_pullup_8v | 20 | Orange | | +8 V to cam pull-up | ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6) | 200 | ST185-AB-cabin | CAM |

### R-CRUISE - resistor r_cruise on CSB3

inline resistor 1k 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_cru_5v_ac | 20 | Orange | | +5 V cruise ladder | DT06-8 plug (ECU splice point) pin 2 | 680 | ST185-AB-cabin | 5V |

### R-FUELLVL - 470R fuel level pull-up to +5V

inline resistor 470R 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w44 | 20 | Orange | | +5 V fuel lvl pullup | DT06-8 plug (ECU splice point) pin 4 | 200 | ST185-AB-cabin | FLVL |

### APS-PEDAL - accelerator pedal sensor

housing TBD | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w210 | 20 | Red | | +5 V pedal sensor 2 | DT06-12 plug (ECU splice point) pin 9 | 1200 | ST185-AB-cabin | 5V |
| **2** | w_aps_gnd2 | 20 | Yellow | | Pedal 2 ground | DT06-12 plug (ECU splice point) pin 8 | 1200 | ST185-AB-cabin | SGND |
| **4** | w_aps_vc1 | 20 | Red | | +5 V pedal sensor 1 | DT06-12 plug (ECU splice point) pin 8 | 1200 | ST185-AB-cabin | 5V |
| **5** | w40 | 20 | Yellow | | Pedal 1 ground | DT06-12 plug (ECU splice point) pin 7 | 1200 | ST185-AB-cabin | SGND |

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ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar)  (part 2 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 6 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar)  (part 3 of 3)

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### IX-FUEL-P - IX-FUEL inline to fuel sender lead (trunk end)

housing DT04-2P  + W2P  |  contacts 0460-202-1631  |  pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w45 | 20 | Blue | | Fuel level ground | DT06-12 plug (ECU splice point) pin 2 | 2820 | ST185-AB-cabin | SGND |

### IX-WS-F-S - to front VR conditioner

housing DT06-6S  + W6S  |  contacts 0462-201-1631  |  pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_fspur_5v | 20 | Red | | VRC-F +5V | DT06-12 plug (ECU splice point) pin 10 | 640 | ST185-AB-cabin | 5V |
| **2** | cab_fspur_gnd | 20 | Yellow | | VRC-F GND | DT06-12 plug (ECU splice point) pin 9 | 640 | ST185-AB-cabin | SGND |

### IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)

housing DT06-6S  + W6S  |  contacts 0462-201-1631  |  pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_rspur_5v | 20 | Red | | VRC-R +5V | DT06-12 plug (ECU splice point) pin 11 | 640 | ST185-AB-cabin | 5V |
| **2** | cab_rspur_gnd | 20 | Yellow | | VRC-R GND | DT06-12 plug (ECU splice point) pin 3 | 640 | ST185-AB-cabin | SGND |

### POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_csbg_ac | 20 | Black | | CSB3 + opto ground | CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground | 1050 | ST185-AB-cabin | CSB |
| | w66 | 10 | Black | | ECU power ground | ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar | 570 | ST185-AB-cabin | GND |
| | w_ac_cool_gnd | 20 | Black | | A/C coolant-OK ground | OEM A/C amplifier A18 (1993 book: A15) pin 2 | 860 | ST185-AB-cabin | ACCGND |

### OEM A/C amplifier A18 (1993 book: A15)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_ac_cool_gnd | 20 | Black | | A/C coolant-OK ground | POWER box NEG bar M8 stud (glove box) | 860 | ST185-AB-cabin | ACCGND |

### OEM J/B No.1 (LH kick panel) 1H pin 7, B-O

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **7** | w_ig2_in | 20 | Yellow | | IG2 wake feed | IG2 wake diode (1N5822) pin 1 | 1210 | ST185-AB-cabin | IG2WAKE |

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| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 7 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 1 of 5)

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### ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **9** | w_trig2_pin | 20 | White | | Cam sensor signal | At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V | 310 | ST185-AB-cabin | CAM |
| **14** | w38 | 20 | Blue | | Pedal 1 signal main | accelerator pedal sensor pin 6 | 1090 | ST185-AB-cabin | APS |
| **18** | w_ac_kill | 20 | White | | A/C cancel (ACT) | OEM A/C amplifier pin 11 | 820 | ST185-AB-cabin | ACKILL |
| **19** | w_fp85 | 18 | White | | Fuel pump relay (-) | B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ... | 3130 | ST185-AB-cabin | FP |
| **23** | cab_fspur_l | 20 | Blue | | FL conditioned out | to front VR conditioner pin 3 | 530 | ST185-AB-cabin | WSSF |
| **26** | | | | | spare, not used | | | | |
| **29** | w_fan85 | 18 | White | | Rad fan relay (-) | B1-R4 Radiator fan relay (seated in the POWER box R4) pin 85 | 680 | ST185-AB-cabin | FAN |

### ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts TBD | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | | | | | spare, not used | | | | |
| **2** | | | | | spare, not used | | | | |
| **3** | | | | | spare, not used | | | | |
| **4** | | | | | spare, not used | | | | |
| **8** | | | | | spare, not used | | | | |
| **10** | w_tach | 20 | White | | Engine speed (tacho) | SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap | 820 | ST185-AB-cabin | TACH |
| **11** | | | | | spare, not used | | | | |
| **12** | w_eps_trig | 18 | White | | EPS enable relay (-) | B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85... | 680 | ST185-AB-cabin | EPSEN |
| **13** | w_fan285 | 18 | White | | Cond fan relay (-) | B1-R5 Condenser fan relay (seated in the POWER box R5) pin 85 | 680 | ST185-AB-cabin | FAN2 |
| **14** | | | | | spare, not used | | | | |
| **19** | cab_rspur_r | 20 | Blue | | Rear right wheel spd | IX-WS-R inline to rear VRC (dash above ECU) pin 4 | 530 | ST185-AB-cabin | WSSR |
| **20** | cab_rspur_l | 20 | Green | | Rear left wheel spd | IX-WS-R inline to rear VRC (dash above ECU) pin 3 | 530 | ST185-AB-cabin | WSSR |
| **21** | cab_fspur_r_ac | 20 | White | | FR conditioned out | to front VR conditioner pin 4 | 530 | ST185-AB-cabin | WSSF |
| **24** | w_fl_sig_pin | 20 | White | | Fuel level signal | SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24 | 310 | ST185-AB-cabin | FLVL |
| **28** | w_ign | 20 | Yellow | | Ignition sw sense | OEM J/B No.1 (LH kick panel) 1H pin 7, B-O pin 7 | 1100 | ST185-AB-cabin | IGNSW |
| **30** | w_start | 20 | Yellow | | Start request | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 680 | ST185-AB-cabin | B30 |
| **31** | | | | | spare, not used | | | | |
| **32** | | | | | spare, not used | | | | |
| **33** | w_aps_vpa2 | 20 | Blue | | Pedal 2 signal sub | accelerator pedal sensor pin 3 | 1090 | ST185-AB-cabin | APS |

### BUS-5V-2 - DT06-8 plug (ECU splice point)

housing DT06-08SA  + W8S; DT04-08PA-P021 (bussed cap) | contacts 114017 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | | | | | spare, sealing plug | | | | |
| **6** | | | | | spare, sealing plug | | | | |
| **7** | | | | | spare, sealing plug | | | | |
| **8** | | | | | spare, sealing plug | | | | |

### BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA  + W12S; DT04-12PA-P021 (bussed cap) | contacts 114017 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **10** | | | | | spare, sealing plug | | | | |
| **11** | | | | | spare, sealing plug | | | | |
| **12** | | | | | spare, sealing plug | | | | |

### BUS-GNDOUT-B - DT06-12 plug (ECU splice point)

housing DT06-12SA  + W12S; DT04-12PA-P021 (bussed cap) | contacts 114017 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **9** | | | | | spare, sealing plug | | | | |
| **10** | | | | | spare, sealing plug | | | | |
| **11** | | | | | spare, sealing plug | | | | |
| **12** | | | | | spare, sealing plug | | | | |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 1 of 5)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 8 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 2 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-AC-PSW - At A/C amp - SP-AC-PSW: A/C pressure switch output

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_ac_psw_in | 20 | Yellow | | A/C pressure sw out | OEM A5, factory Y-R lead pin 4 | 200 | ST185-AB-cabin | ACC |
| | w_ac_psw_sw | 20 | Yellow | | A/C pressure sw out | OEM A/C amplifier pin 13 | 200 | ST185-AB-cabin | ACC |

### SP-AC-SG - cabin A/C sensor common (replaces E10; A32-15, A24-3, A21, A18-2 stay factory)

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_amb_rtn | 20 | Yellow | | Ambient sensor return | OEM ambient air temp sensor A1 pin 1 (front, 6 in pigtail) pin 1 | 2010 | ST185-AB-cabin | AMB |
| | w_amb_sg | 20 | Yellow | | A/C sensor common | OEM auto A/C amplifier A32 pin 15 SG pin 15 | 200 | ST185-AB-cabin | AMB |

### SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_acc86 | 18 | Red | | A/C clutch relay coil+ | B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B... | 200 | ST185-AB-cabin | ACC |

### SP-CAM - At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_trig2_pin | 20 | White | | Cam sensor signal | Link G4X plug A (centre dash) pin 9 | 310 | ST185-AB-cabin | CAM |
| | w_cam_pullup_sig | 20 | White | | Cam pull-up 1.8k | Cam pull-up 1.8 k to +8 V pin 2 | 200 | ST185-AB-cabin | CAM |

### SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fl_sig | 20 | White | | Fuel level signal | IX-FUEL inline to fuel sender lead (trunk end) pin 1 | 2820 | ST185-AB-cabin | FLVL |
| | w_fl_sig_pin | 20 | White | | Fuel level signal | ECU-B (Link G4X plug B, centre dash) pin 24 | 310 | ST185-AB-cabin | FLVL |
| | w43 | 20 | White | | Fuel lvl pull-up | 470R fuel level pull-up to +5V pin 1 | 200 | ST185-AB-cabin | FLVL |

### SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_sw7_c | 16 | Yellow | | Reverse to opto in 2 | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin IN2 | 680 | ST185-AB-cabin | REV |
| | w_rev_lamps | 16 | Yellow | | Back-up lamp +12 V | OEM IG2 pin 13 (under the ECU, cowl side) - factory back-up lamp output pin 13 | 310 | ST185-AB-cabin | REV |

### SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_tach | 20 | White | | Engine speed (tacho) | ECU-B (Link G4X plug B, centre dash) pin 10 | 820 | ST185-AB-cabin | TACH |
| | w_tach_18 | 20 | White | | Engine speed (tacho) | OEM A/C amplifier A18 (1993 book: A15) pin 18 | 200 | ST185-AB-cabin | TACH |
| | w_tach_r | 20 | White | | Tacho to lock-sensor 4.7k | Lock-sensor tap 4.7 k from tacho pin 1 | 200 | ST185-AB-cabin | TACH |

### R-CAM-PULLUP - Cam pull-up 1.8 k to +8 V

inline resistor 1.8k 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_cam_pullup_sig | 20 | White | | Cam pull-up 1.8k | At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V | 200 | ST185-AB-cabin | CAM |

### R-FUELLVL - 470R fuel level pull-up to +5V

inline resistor 470R 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w43 | 20 | White | | Fuel lvl pull-up | SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24 | 200 | ST185-AB-cabin | FLVL |

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| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
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**ST185-harness - every wiring harness of the car, one drawing set**

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 2 of 5)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 9 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 3 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### R-LOCK - Lock-sensor tap 4.7 k from tacho

inline resistor 4.7k 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_tach_r | 20 | White | | Tacho to lock-sensor 4.7k | SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap | 200 | ST185-AB-cabin | TACH |
| **2** | w_lock_9 | 20 | White | | Lock-sensor input (modified tacho) | OEM A/C amplifier A18 (1993 book: A15) pin 9 | 200 | ST185-AB-cabin | TACH |

### DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018 + DRBF-1A | contacts 0413-214-1205 | pins per TE drawing DRB12-102PXE-L018 rev C3

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | | | | | spare, sealing plug | | | | |
| **6** | | | | | spare, sealing plug | | | | |
| **7** | | | | | spare, sealing plug | | | | |
| **8** | | | | | spare, sealing plug | | | | |
| **19** | | | | | spare, sealing plug | | | | |
| **20** | | | | | spare, sealing plug | | | | |
| **35** | | | | | spare, sealing plug | | | | |
| **36** | | | | | spare, sealing plug | | | | |
| **37** | | | | | spare, sealing plug | | | | |
| **38** | | | | | spare, sealing plug | | | | |
| **39** | | | | | spare, sealing plug | | | | |
| **40** | | | | | spare, sealing plug | | | | |
| **43** | | | | | spare, sealing plug | | | | |
| **48** | | | | | spare, sealing plug | | | | |
| **49** | | | | | spare, sealing plug | | | | |
| **51** | | | | | spare, sealing plug | | | | |
| **54** | | | | | spare, sealing plug | | | | |
| **55** | | | | | spare, sealing plug | | | | |
| **60** | | | | | spare, sealing plug | | | | |
| **62** | | | | | spare, sealing plug | | | | |
| **82** | | | | | spare, sealing plug | | | | |
| **86** | | | | | spare, sealing plug | | | | |
| **87** | | | | | spare, sealing plug | | | | |
| **88** | | | | | spare, sealing plug | | | | |
| **89** | | | | | spare, sealing plug | | | | |
| **90** | | | | | spare, sealing plug | | | | |
| **91** | | | | | spare, sealing plug | | | | |
| **92** | | | | | spare, sealing plug | | | | |
| **93** | | | | | spare, sealing plug | | | | |
| **94** | | | | | spare, sealing plug | | | | |
| **97** | | | | | spare, sealing plug | | | | |
| **98** | | | | | spare, sealing plug | | | | |
| **99** | | | | | spare, sealing plug | | | | |
| **100** | | | | | spare, sealing plug | | | | |
| **101** | | | | | spare, sealing plug | | | | |
| **102** | | | | | spare, sealing plug | | | | |

### APS-PEDAL - accelerator pedal sensor

housing TBD | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | w_aps_vpa2 | 20 | Blue | | Pedal 2 signal sub | ECU-B (Link G4X plug B, centre dash) pin 33 | 1090 | ST185-AB-cabin | APS |
| **6** | w38 | 20 | Blue | | Pedal 1 signal main | Link G4X plug A (centre dash) pin 14 | 1090 | ST185-AB-cabin | APS |

### IX-FUEL-P - IX-FUEL inline to fuel sender lead (trunk end)

housing DT04-2P + W2P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_fl_sig | 20 | White | | Fuel level signal | SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24 | 2820 | ST185-AB-cabin | FLVL |

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CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 3 of 5)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 10 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 4 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### IX-WS-F-S - to front VR conditioner

housing DT06-6S  + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | cab_fspur_l | 20 | Blue | | FL conditioned out | Link G4X plug A (centre dash) pin 23 | 530 | ST185-AB-cabin | WSSF |
| **4** | cab_fspur_r_ac | 20 | White | | FR conditioned out | ECU-B (Link G4X plug B, centre dash) pin 21 | 530 | ST185-AB-cabin | WSSF |
| **6** | | | | | spare, sealing plug | | | | |

### IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)

housing DT06-6S  + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | cab_rspur_l | 20 | Green | | Rear left wheel spd | ECU-B (Link G4X plug B, centre dash) pin 20 | 530 | ST185-AB-cabin | WSSR |
| **4** | cab_rspur_r | 20 | Blue | | Rear right wheel spd | ECU-B (Link G4X plug B, centre dash) pin 19 | 530 | ST185-AB-cabin | WSSR |
| **6** | | | | | spare, sealing plug | | | | |

### K-ACC - B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B2-BUS2-8

housing 6-1419137-4 | contacts 280919-2 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **85** | w_acc_trig | 18 | White | | A/C clutch relay coil (-) | OEM A/C amplifier A18 (1993 book: A15) pin 8 | 860 | ST185-AB-cabin | ACC |
| **86** | w_acc86 | 18 | Red | | A/C clutch relay coil+ | SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86 | 200 | ST185-AB-cabin | ACC |

### K-EPS-EN - B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V

housing 6-1419137-4 | contacts 280919-2 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **85** | w_eps_trig | 18 | White | | EPS enable relay (-) | ECU-B (Link G4X plug B, centre dash) pin 12 | 680 | ST185-AB-cabin | EPSEN |

### K-FAN - B1-R4 Radiator fan relay (seated in the POWER box R4)

housing 1-1414147-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **85** | w_fan85 | 18 | White | | Rad fan relay (-) | Link G4X plug A (centre dash) pin 29 | 680 | ST185-AB-cabin | FAN |

### K-FAN2 - B1-R5 Condenser fan relay (seated in the POWER box R5)

housing 1-1393304-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **85** | w_fan285 | 18 | White | | Cond fan relay (-) | ECU-B (Link G4X plug B, centre dash) pin 13 | 680 | ST185-AB-cabin | FAN2 |

### K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts 282110-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **85** | w_fp85 | 18 | White | | Fuel pump relay (-) | Link G4X plug A (centre dash) pin 19 | 3130 | ST185-AB-cabin | FP |

### LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968849-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **BUS2-13** | w_start | 20 | Yellow | | Start request | ECU-B (Link G4X plug B, centre dash) pin 30 | 680 | ST185-AB-cabin | B30 |

### OPTO-8 - Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3

housing TBD | contacts screw terminal (on the board) | pins per terminal names printed on the board

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **IN2** | w_sw7_c | 16 | Yellow | | Reverse to opto in 2 | SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps | 680 | ST185-AB-cabin | REV |

### OEM A/C amplifier

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **11** | w_ac_kill | 20 | White | | A/C cancel (ACT) | Link G4X plug A (centre dash) pin 18 | 820 | ST185-AB-cabin | ACKILL |
| **13** | w_ac_psw_sw | 20 | Yellow | | A/C pressure sw out | At A/C amp - SP-AC-PSW: A/C pressure switch output | 200 | ST185-AB-cabin | ACC |

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| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 4 of 5)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 11 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 5 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### OEM A/C amplifier A18 (1993 book: A15)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **18** | w_tach_18 | 20 | White | | Engine speed (tacho) | SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap | 200 | ST185-AB-cabin | TACH |
| **8** | w_acc_trig | 18 | White | | A/C clutch relay coil (-) | B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B... | 860 | ST185-AB-cabin | ACC |
| **9** | w_lock_9 | 20 | White | | Lock-sensor input (modified tacho) | Lock-sensor tap 4.7 k from tacho pin 2 | 200 | ST185-AB-cabin | TACH |

### OEM ambient air temp sensor A1 pin 1 (front, 6 in pigtail)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_amb_rtn | 20 | Yellow | | Ambient sensor return | SP-AC-SG - cabin A/C sensor common (replaces E10; A32-15, A24-3, A21, A18-2 stay factory) | 2010 | ST185-AB-cabin | AMB |

### OEM ambient air temp sensor A1 pin 2 (front, 6 in pigtail)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_amb_sig | 20 | Yellow | | Ambient temp signal | OEM auto A/C amplifier A32 pin 6 TAM pin 6 | 2010 | ST185-AB-cabin | AMB |

### OEM auto A/C amplifier A32 pin 15 SG

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **15** | w_amb_sg | 20 | Yellow | | A/C sensor common | SP-AC-SG - cabin A/C sensor common (replaces E10; A32-15, A24-3, A21, A18-2 stay factory) | 200 | ST185-AB-cabin | AMB |

### OEM auto A/C amplifier A32 pin 6 TAM

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **6** | w_amb_sig | 20 | Yellow | | Ambient temp signal | OEM ambient air temp sensor A1 pin 2 (front, 6 in pigtail) pin 2 | 2010 | ST185-AB-cabin | AMB |

### OEM A5, factory Y-R lead

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_ac_psw_in | 20 | Yellow | | A/C pressure sw out | At A/C amp - SP-AC-PSW: A/C pressure switch output | 200 | ST185-AB-cabin | ACC |

### OEM IG2 pin 13 (under the ECU, cowl side) - factory back-up lamp output

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **13** | w_rev_lamps | 16 | Yellow | | Back-up lamp +12 V | SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps | 310 | ST185-AB-cabin | REV |

### OEM J/B No.1 (LH kick panel) 1H pin 7, B-O

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **7** | w_ign | 20 | Yellow | | Ignition sw sense | ECU-B (Link G4X plug B, centre dash) pin 28 | 1100 | ST185-AB-cabin | IGNSW |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses  (part 5 of 5)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 12 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk  (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **7** | w_drain_ecu_a | 20 | Black | | Shield drain | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 200 | ST185-AB-cabin | SHLD |

### ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **17** | w_drain_ecu_b | 20 | Black | | Shield drain B | 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17 | 200 | ST185-AB-cabin | SHLD |

### SP-CAN-SH-BH - CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_canbh_bc_sh | | Shield | | CAN-Lambda shield | DRB102 receptacle (RH firewall) pin 81 | 680 | ST185-AB-cabin | CAN |

### SP-SHIELD-A - 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_drain_ecu_a | 20 | Black | | Shield drain | Link G4X plug A (centre dash) pin 7 | 200 | ST185-AB-cabin | SHLD |
| | w_shd_a_aps | 20 | Black | | Shield drain | At the pedal branch - SP-SHIELD-A-APS: pedal drains | 450 | ST185-AB-cabin | SHLD |
| | w_shd_a_wsf | 20 | Black | | Shield drain | At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains | 230 | ST185-AB-cabin | SHLD |
| | w_shc_crank | 20 | Black | | Crank sensor shield | DRB102 receptacle (RH firewall) pin 76 | 630 | ST185-AB-cabin | CRANKC |
| | w_shc_cam | 20 | Black | | Cam sensor shield | DRB102 receptacle (RH firewall) pin 66 | 630 | ST185-AB-cabin | CAM |
| | w_shc_turbo | 20 | Black | | Turbo speed shield | DRB102 receptacle (RH firewall) pin 57 | 630 | ST185-AB-cabin | TURBO |
| | w_shc_flex | 20 | Black | | Flex fuel shield | DRB102 receptacle (RH firewall) pin 32 | 630 | ST185-AB-cabin | FLEX |
| | w_shc_etbs | 20 | Black | | Throttle shield | DRB102 receptacle (RH firewall) pin 47 | 630 | ST185-AB-cabin | ETBS |

### SP-SHIELD-A-APS - At the pedal branch - SP-SHIELD-A-APS: pedal drains

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_shd_a_aps | 20 | Black | | Shield drain | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 450 | ST185-AB-cabin | SHLD |
| | cab_aps1_sh | | Shield | | Pedal 1 cable shield | cab_aps1 screen (floats at the device end) | 840 | ST185-AB-cabin | APS |
| | cab_aps_sh | | Shield | | Pedal 2 cable shield | cab_aps2 screen (floats at the device end) | 840 | ST185-AB-cabin | APS |

### SP-SHIELD-A-WSF - At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_shd_a_wsf | 20 | Black | | Shield drain | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 230 | ST185-AB-cabin | SHLD |
| | cab_fspur_sh | | Shield | | WS-F screen drain | to front VR conditioner pin 5 | 500 | ST185-AB-cabin | WSSF |
| | cab_fspur_fr_sh_ac | | Shield | | FR wheel spd shield | cab_fspur_fr_ac screen (floats at the device end) | 500 | ST185-AB-cabin | WSSF |

### SP-SHIELD-B - 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_shc_knock1 | 20 | Black | | Knock sensor shield | DRB102 receptacle (RH firewall) pin 73 | 630 | ST185-AB-cabin | KNOCK |
| | w_shc_etbm | 20 | Black | | ETB motor shield | DRB102 receptacle (RH firewall) pin 50 | 630 | ST185-AB-cabin | ETBM |
| | w_shc_vss | 20 | Black | | Gearbox speed shield | DRB102 receptacle (RH firewall) pin 69 | 630 | ST185-AB-cabin | VSS |
| | w_drain_ecu_b | 20 | Black | | Shield drain B | ECU-B (Link G4X plug B, centre dash) pin 17 | 200 | ST185-AB-cabin | SHLD |
| | w_shd_b_wsr | 20 | Black | | Shield drain B | At the rear WS inline - SP-SHIELD-B-WSR: rear wheel-speed drains | 230 | ST185-AB-cabin | SHLD |

### SP-SHIELD-B-WSR - At the rear WS inline - SP-SHIELD-B-WSR: rear wheel-speed drains

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_rspur_sh | | Shield | | WS-R screen drain | IX-WS-R inline to rear VRC (dash above ECU) pin 5 | 500 | ST185-AB-cabin | WSSR |
| | w_shd_b_wsr | 20 | Black | | Shield drain B | 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17 | 230 | ST185-AB-cabin | SHLD |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
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| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
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**ST185-harness - every wiring harness of the car, one drawing set**

SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk  (part 1 of 2)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 13 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

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## SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk  (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018  + DRBF-1A  |  contacts 0460-202-1631  |  pins per TE drawing DRB12-102PXE-L018 rev C3

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **32** | w_shc_flex | 20 | Black | | Flex fuel shield | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 630 | ST185-AB-cabin | FLEX |
| **47** | w_shc_etbs | 20 | Black | | Throttle shield | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 630 | ST185-AB-cabin | ETBS |
| **50** | w_shc_etbm | 20 | Black | | ETB motor shield | 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17 | 630 | ST185-AB-cabin | ETBM |
| **57** | w_shc_turbo | 20 | Black | | Turbo speed shield | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 630 | ST185-AB-cabin | TURBO |
| **66** | w_shc_cam | 20 | Black | | Cam sensor shield | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 630 | ST185-AB-cabin | CAM |
| **69** | w_shc_vss | 20 | Black | | Gearbox speed shield | 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17 | 630 | ST185-AB-cabin | VSS |
| **73** | w_shc_knock1 | 20 | Black | | Knock sensor shield | 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17 | 630 | ST185-AB-cabin | KNOCK |
| **76** | w_shc_crank | 20 | Black | | Crank sensor shield | 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7 | 630 | ST185-AB-cabin | CRANKC |
| **81** | cab_canbh_bc_sh | | Shield | | CAN-Lambda shield | CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda | 680 | ST185-AB-cabin | CAN |

### IX-WS-F-S - to front VR conditioner

housing DT06-6S  + W6S  |  contacts 0462-201-1631  |  pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | cab_fspur_sh | | Shield | | WS-F screen drain | At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains | 500 | ST185-AB-cabin | WSSF |

### IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)

housing DT06-6S  + W6S  |  contacts 0462-201-1631  |  pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | cab_rspur_sh | | Shield | | WS-R screen drain | At the rear WS inline - SP-SHIELD-B-WSR: rear wheel-speed drains | 500 | ST185-AB-cabin | WSSR |

### CAB-APS1 cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_aps1_sh | | Shield | | Pedal 1 cable shield | At the pedal branch - SP-SHIELD-A-APS: pedal drains | 840 | ST185-AB-cabin | APS |

### CAB-APS2 cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_aps_sh | | Shield | | Pedal 2 cable shield | At the pedal branch - SP-SHIELD-A-APS: pedal drains | 840 | ST185-AB-cabin | APS |

### CAB-FSPUR-FR-AC cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_fspur_fr_sh_ac | | Shield | | FR wheel spd shield | At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains | 500 | ST185-AB-cabin | WSSF |

**REVISIONS**

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SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk  (part 2 of 2)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 14 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 1 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-CANH-LAM - CAN H join - bulkhead, Lambda, 120 ohm end

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_h_lam_be | 20 | White | | CAN H to CAN-Lambda | BULKHEAD pin 79 | 1157 | ST185-AB-engine | CAN |
| | wc_term3 | 20 | Green | | Lambda 120R CAN H | r_term_end pin 1 | 200 | ST185-AB-engine | CAN |
| | wc_h_lam_pin | 20 | White | | CAN H to CAN-Lambda | Link CAN-Lambda pin 4 | 200 | ST185-AB-engine | CAN |

### SP-CANL-LAM - CAN L join - bulkhead, Lambda, 120 ohm end

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_l_lam_be | 20 | Green | | CAN L to CAN-Lambda | BULKHEAD pin 80 | 1157 | ST185-AB-engine | CAN |
| | wc_term4 | 20 | Green | | Lambda 120R CAN L | r_term_end pin 2 | 200 | ST185-AB-engine | CAN |
| | wc_l_lam_pin | 20 | Green | | CAN L to CAN-Lambda | Link CAN-Lambda pin 3 | 200 | ST185-AB-engine | CAN |

### SP-CHASSIS-ENG - Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar)

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_cop1g | 18 | Black | | Coil 1 ground | COP 1 (1ZZ) pin 4 | 493 | ST185-AB-engine | COP1 |
| | w_cop2g | 18 | Black | | Coil 2 ground | COP 2 (1ZZ) pin 4 | 560 | ST185-AB-engine | COP2 |
| | w_cop3g | 18 | Black | | Coil 3 ground | COP 3 (1ZZ) pin 4 | 457 | ST185-AB-engine | COP3 |
| | w_cop4g | 18 | Black | | Coil 4 ground | COP 4 (1ZZ) pin 4 | 352 | ST185-AB-engine | COP4 |
| | w_cop_gnd_e_ae | 12 | Black | | Coil ground | BULKHEAD pin 4 | 1198 | ST185-AB-engine | COPG |

### SP-COP-ENG - Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_cop_pwr_e_ae | 12 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | BULKHEAD pin 1 | 1198 | ST185-AB-engine | COP |
| | w112 | 16 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | COP 1 (1ZZ) pin 1 | 493 | ST185-AB-engine | COP |
| | w113 | 16 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | COP 2 (1ZZ) pin 1 | 560 | ST185-AB-engine | COP |
| | w114 | 16 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | COP 3 (1ZZ) pin 1 | 457 | ST185-AB-engine | COP |
| | w115 | 16 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | COP 4 (1ZZ) pin 1 | 352 | ST185-AB-engine | COP |

### SP-GNDOUT-B-ENG - Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w10 | 20 | Blue | | Gnd charge temp | Charge-Pipe IAT Sensor pin 2 | 1690 | ST185-AB-engine | CPIAT |
| | w8 | 20 | Blue | | Gnd oil temp | Oil Temp Sensor pin 2 | 1391 | ST185-AB-engine | OILT |
| | w_rail_gnd_b_e | 20 | Blue | | Gnd oil/charge temp | Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure | 200 | ST185-AB-engine | GNDOUT |

### SP-GNDOUT-B-ENG2 - Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_b_e_c19 | 20 | Blue | | Gnd oil/clnt/temps | BULKHEAD pin 29 | 403 | ST185-AB-engine | GNDOUT |
| | w36 | 20 | Blue | | Gnd coolant press | Coolant Pressure Sensor pin 1 | 810 | ST185-AB-engine | CLNTP |
| | w30 | 20 | Blue | | Gnd oil pressure | Oil Pressure Sensor pin 1 | 2045 | ST185-AB-engine | OILP |
| | w_rail_gnd_b_e | 20 | Blue | | Gnd oil/charge temp | Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice | 200 | ST185-AB-engine | GNDOUT |

### SP-GNDOUT-B-ENG3 - Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_b_e_c21 | 20 | Blue | | Gnd fuel press + VSS | BULKHEAD pin 15 | 1094 | ST185-AB-engine | GNDOUT |
| | w33 | 20 | Blue | | Gnd fuel pressure | Fuel Pressure Sensor pin 1 | 610 | ST185-AB-engine | FUELP |
| | w_vss_gnd | 20 | Yellow | | Gnd gearbox speed | Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143 pin 2 | 684 | ST185-AB-engine | VSS |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
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**ST185-harness - every wiring harness of the car, one drawing set**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 1 of 8)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 15 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 2 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-GNDOUT-ENG - Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle

splice: crimp splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rail_sp_gndout_e | 20 | Blue | | Gnd IAT/TPS/coolant | BULKHEAD pin 34 | 403 | ST185-AB-engine | GNDOUT |
| | w4 | 20 | Blue | | Gnd coolant temp | ECT Sensor pin 2 | 1812 | ST185-AB-engine | ECT |
| | w127 | 20 | Yellow | | Gnd throttle sensor | Bosch 74.5mm ETB pin 2 | 1716 | ST185-AB-engine | ETBS |
| | w6 | 20 | Blue | | Gnd intake temp | IAT Sensor (Manifold) pin 2 | 1285 | ST185-AB-engine | IAT |

### SP-GNDOUT-ENG2 - Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_a_e_c44 | 20 | Blue | | Gnd flex + MAP | BULKHEAD pin 18 | 1270 | ST185-AB-engine | GNDOUT |
| | w47 | 20 | Yellow | | Gnd flex fuel | Flex Fuel Sensor pin B | 315 | ST185-AB-engine | FLEX |
| | w27 | 20 | Blue | | Gnd MAP sensor | MAP Sensor pin 1 | 356 | ST185-AB-engine | MAP |

### SP-INJ-ENG - Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_inj_pwr_e_ae | 12 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | BULKHEAD pin 2 | 980 | ST185-AB-engine | INJ |
| | w211 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector 1 pin 1 | 475 | ST185-AB-engine | INJ |
| | w212 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector 2 pin 1 | 388 | ST185-AB-engine | INJ |
| | w213 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector 3 pin 1 | 307 | ST185-AB-engine | INJ |
| | w214 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector 4 pin 1 | 242 | ST185-AB-engine | INJ |

### SP-LAM12-B - CAN-Lambda +12 V splice - 12 AWG from BH-B pin 95 to the 16 AWG lead

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_lam12_be14 | 12 | Red | | EFI relay k_efi -> LOGIC LAM 7.5 A fuse | BULKHEAD pin 95 | 1157 | ST185-AB-engine | LAM |
| | w_lam12_be | 16 | Red | | EFI relay k_efi -> LOGIC LAM 7.5 A fuse | Link CAN-Lambda pin 1 | 200 | ST185-AB-engine | LAM |

### SP-LAMG-B - CAN-Lambda ground splice - 12 AWG from BH-B pin 96 to the 16 AWG lead

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_lamg_be | 16 | Black | | CAN-Lambda ground | Link CAN-Lambda pin 2 | 200 | ST185-AB-engine | LAM |
| | w_lamg_be14 | 12 | Black | | CAN-Lambda ground | BULKHEAD pin 96 | 1157 | ST185-AB-engine | LAM |

### SP-SW12-ENG - Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rail_sp_sw12_e | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | BULKHEAD pin 30 | 403 | ST185-AB-engine | SW12 |
| | w219 | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | Boost Control Solenoid pin 1 | 1840 | ST185-AB-engine | SW12 |
| | w217 | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | Flex Fuel Sensor pin A | 1182 | ST185-AB-engine | SW12 |

### R-TERM-END - r_term_end

inline resistor 120R 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | wc_term3 | 20 | Green | | Lambda 120R CAN H | CAN H join - bulkhead, Lambda, 120 ohm end | 200 | ST185-AB-engine | CAN |
| **2** | wc_term4 | 20 | Green | | Lambda 120R CAN L | CAN L join - bulkhead, Lambda, 120 ohm end | 200 | ST185-AB-engine | CAN |

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**ST185-harness - every wiring harness of the car, one drawing set**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 2 of 8)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 16 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 3 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### DRB102-ENGINE - BULKHEAD

housing DRB16-102SAE-L018 | contacts 0462-210-1231 | pins per TE drawing DRB12-102PXE-L018 rev C3

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_cop_pwr_e_ae | 12 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4 | 1198 | ST185-AB-engine | COP |
| 2 | w_inj_pwr_e_ae | 12 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4 | 980 | ST185-AB-engine | INJ |
| 3 | w_strl_e | 12 | Red | | Starter solenoid S | Starter solenoid S (50) terminal post, spade | 2433 | ST185-AB-engine | STR |
| 4 | w_cop_gnd_e_ae | 12 | Black | | Coil ground to the POWER box NEG bar | Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar) | 1198 | ST185-AB-engine | COPG |
| 5 | | | | | spare, sealing plug | | | | |
| 6 | | | | | spare, sealing plug | | | | |
| 7 | | | | | spare, sealing plug | | | | |
| 8 | | | | | spare, sealing plug | | | | |
| 9 | w120_e | 20 | White | | Coil 1 trigger (IGT) | COP 1 (1ZZ) pin 3 | 1491 | ST185-AB-engine | COP1 |
| 10 | w121_e | 20 | White | | Coil 2 trigger (IGT) | COP 2 (1ZZ) pin 3 | 1558 | ST185-AB-engine | COP2 |
| 11 | w122_e | 20 | White | | Coil 3 trigger (IGT) | COP 3 (1ZZ) pin 3 | 1455 | ST185-AB-engine | COP3 |
| 12 | w123_e | 20 | White | | Coil 4 trigger (IGT) | COP 4 (1ZZ) pin 3 | 1350 | ST185-AB-engine | COP4 |
| 13 | w_5v_brail_e_c46 | 20 | Orange | | +5 V fuel pressure | Fuel Pressure Sensor pin 2 | 1504 | ST185-AB-engine | 5V |
| 14 | w32_e | 20 | White | | Fuel pressure signal | Fuel Pressure Sensor pin 3 | 1504 | ST185-AB-engine | FUELP |
| 15 | w_gnd_b_e_c21 | 20 | Blue | | Gnd fuel press + VSS | Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed | 1094 | ST185-AB-engine | GNDOUT |
| 16 | w206 | 20 | Orange | | +5 V MAP sensor | MAP Sensor pin 2 | 1426 | ST185-AB-engine | MAP |
| 17 | w26_e | 20 | White | | MAP sensor signal | MAP Sensor pin 3 | 1426 | ST185-AB-engine | MAP |
| 18 | w_gnd_a_e_c44 | 20 | Blue | | Gnd flex + MAP | Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor | 1270 | ST185-AB-engine | GNDOUT |
| 19 | | | | | spare, sealing plug | | | | |
| 20 | | | | | spare, sealing plug | | | | |
| 21 | w108_e | 20 | White | | Injector 1 drive (-) | Injector 1 pin 2 | 1255 | ST185-AB-engine | INJ1 |
| 22 | w109_e | 20 | White | | Injector 2 drive (-) | Injector 2 pin 2 | 1168 | ST185-AB-engine | INJ2 |
| 23 | w110_e | 20 | White | | Injector 3 drive (-) | Injector 3 pin 2 | 1087 | ST185-AB-engine | INJ3 |
| 24 | w111_e | 20 | White | | Injector 4 drive (-) | Injector 4 pin 2 | 1022 | ST185-AB-engine | INJ4 |
| 25 | w_5v_brail_e_c47 | 20 | Orange | | +5 V coolant press | Coolant Pressure Sensor pin 2 | 1013 | ST185-AB-engine | 5V |
| 26 | w35_e | 20 | White | | Coolant press signal | Coolant Pressure Sensor pin 3 | 1013 | ST185-AB-engine | CLNTP |
| 27 | w_5v_brail_e_c45 | 20 | Orange | | +5 V oil pressure | Oil Pressure Sensor pin 2 | 2248 | ST185-AB-engine | 5V |
| 28 | w29_e | 20 | White | | Oil pressure signal | Oil Pressure Sensor pin 3 | 2248 | ST185-AB-engine | OILP |
| 29 | w_gnd_b_e_c19 | 20 | Blue | | Gnd oil/clnt/temps | Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure | 403 | ST185-AB-engine | GNDOUT |
| 30 | w_rail_sp_sw12_e | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor | 403 | ST185-AB-engine | SW12 |
| 31 | w48_e | 20 | Blue | | Flex fuel signal | Flex Fuel Sensor pin C | 1385 | ST185-AB-engine | FLEX |
| 32 | cab_flex_sh | | Shield | | Flex fuel shield | cab_flex screen (floats at the device end) | 1385 | ST185-AB-engine | FLEX |
| 33 | w202_e | 20 | White | | Intake temp signal | IAT Sensor (Manifold) pin 1 | 1488 | ST185-AB-engine | IAT |
| 34 | w_rail_sp_gndout_e | 20 | Blue | | Gnd IAT/TPS/coolant | Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle | 403 | ST185-AB-engine | GNDOUT |
| 35 | | | | | spare, sealing plug | | | | |
| 36 | | | | | spare, sealing plug | | | | |
| 37 | | | | | spare, sealing plug | | | | |
| 38 | | | | | spare, sealing plug | | | | |
| 39 | | | | | spare, sealing plug | | | | |
| 40 | | | | | spare, sealing plug | | | | |
| 41 | w126 | 20 | Red | | +5 V throttle sensor | Bosch 74.5mm ETB pin 3 | 1919 | ST185-AB-engine | ETBS |
| 42 | w128_e | 20 | Green | | Throttle pos 1 main | Bosch 74.5mm ETB pin 6 | 1919 | ST185-AB-engine | ETBS |
| 43 | | | | | spare, sealing plug | | | | |
| 44 | w129_e | 20 | Blue | | Throttle pos 2 sub | Bosch 74.5mm ETB pin 5 | 1919 | ST185-AB-engine | ETBS |
| 45 | w124_e_ae | 18 | Red | | ETB motor + | Bosch 74.5mm ETB pin 4 | 1919 | ST185-AB-engine | ETBM |
| 46 | w215_e_ae | 18 | Black | | ETB motor - | Bosch 74.5mm ETB pin 1 | 1919 | ST185-AB-engine | ETBM |
| 47 | cab_etbs_sh | | Shield | | Throttle shield | cab_etbs screen (floats at the device end) | 1919 | ST185-AB-engine | ETBS |
| 48 | | | | | spare, sealing plug | | | | |
| 49 | | | | | spare, sealing plug | | | | |
| 50 | cab_etbm_ae_sh | | Shield | | ETB motor shield | cab_etbm_ae screen (floats at the device end) | 1919 | ST185-AB-engine | ETBM |
| 51 | | | | | spare, sealing plug | | | | |
| 52 | w218 | 20 | Red | | +5 V turbo speed | Turbo Speed Sensor pin A | 2167 | ST185-AB-engine | TURBO |
| 53 | w16_e | 20 | Blue | | Turbo speed signal | Turbo Speed Sensor pin B | 2167 | ST185-AB-engine | TURBO |
| 54 | | | | | spare, sealing plug | | | | |
| 55 | | | | | spare, sealing plug | | | | |
| 56 | w17 | 20 | Yellow | | Gnd turbo speed | Turbo Speed Sensor pin C | 2167 | ST185-AB-engine | TURBO |
| 57 | cab_turbo_sh | | Shield | | Turbo speed shield | cab_turbo screen (floats at the device end) | 2167 | ST185-AB-engine | TURBO |
| 58 | w103_e | 20 | White | | Boost solenoid (-) | Boost Control Solenoid pin 2 | 2043 | ST185-AB-engine | BOOST |
| 59 | w204_e | 20 | White | | Charge temp signal | Charge-Pipe IAT Sensor pin 1 | 1893 | ST185-AB-engine | CPIAT |
| 60 | | | | | spare, sealing plug | | | | |
| 61 | w203_e | 20 | White | | Oil temp signal | Oil Temp Sensor pin 1 | 1594 | ST185-AB-engine | OILT |
| 62 | | | | | spare, sealing plug | | | | |
| 63 | w_rail_sp_8v_e | 20 | Red | | +8 V to cam sensor | Cam Hall Sensor (RacerX) pin 1 | 1152 | ST185-AB-engine | 8V |
| 64 | cab_cam_c2 | 20 | Blue | | Cam sensor signal | Cam Hall Sensor (RacerX) pin 2 | 1152 | ST185-AB-engine | CAM |
| 65 | w24 | 20 | Yellow | | Gnd cam sensor | Cam Hall Sensor (RacerX) pin 3 | 1152 | ST185-AB-engine | CAM |
| 66 | cab_cam_sh | | Shield | | Cam sensor shield | cab_cam screen (floats at the device end) | 1152 | ST185-AB-engine | CAM |
| 67 | w_vss_12v | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143 pin 1 | 1578 | ST185-AB-engine | SW12 |
| 68 | w_vss_sig_e | 20 | Blue | | Gearbox speed signal | Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143 pin 3 | 1578 | ST185-AB-engine | VSS |
| 69 | cab_vss_sh | | Shield | | Gearbox speed shield | cab_vss screen (floats at the device end) | 1578 | ST185-AB-engine | VSS |
| 70 | w201_e | 20 | White | | Coolant temp signal | ECT Sensor pin 1 | 2015 | ST185-AB-engine | ECT |
| 71 | cab_knock1_c1 | 20 | Blue | | Knock sensor signal | Knock Sensor 1 pin 1 | 1051 | ST185-AB-engine | KNOCK1 |
| 72 | cab_knock1_c2 | 20 | White | | Knock sensor (-) | Knock Sensor 1 pin 2 | 1051 | ST185-AB-engine | KNOCK1 |
| 73 | cab_knock1_sh | | Shield | | Knock sensor shield | cab_knock1 screen (floats at the device end) | 1051 | ST185-AB-engine | KNOCK1 |
| 74 | cab_crank_c1 | 20 | White | | Crank sensor signal | Crank VR Sensor (36-2) pin 1 | 1410 | ST185-AB-engine | CRANK |
| 75 | w2 | 20 | Blue | | Gnd crank sensor | Crank VR Sensor (36-2) pin 2 | 1410 | ST185-AB-engine | CRANK |
| 76 | cab_crank_sh | | Shield | | Crank sensor shield | cab_crank screen (floats at the device end) | 1410 | ST185-AB-engine | CRANK |
| 77 | w_rev_feed_e | 16 | Red | | Back-up sw +12 V in | Back-up SW B1 pin 2 | 2341 | ST185-AB-engine | REV |
| 78 | w_sw7_e | 16 | Yellow | | Reverse sw +12 V out | Back-up SW B1 pin 1 | 2341 | ST185-AB-engine | REV |
| 79 | wc_h_lam_be | 20 | White | | CAN H to CAN-Lambda | CAN H join - bulkhead, Lambda, 120 ohm end | 1157 | ST185-AB-engine | CAN |
| 80 | wc_l_lam_be | 20 | Green | | CAN L to CAN-Lambda | CAN L join - bulkhead, Lambda, 120 ohm end | 1157 | ST185-AB-engine | CAN |
| 81 | cab_lamcan_be_sh | | Shield | | CAN-Lambda shield | cab_can_lam screen (floats at the device end) | 1157 | ST185-AB-engine | CAN |
| 82 | | | | | spare, sealing plug | | | | |
| 83 | w_hv_mrs_en_e | 20 | Red | | EFI relay k_efi -> LOGIC MRS 7.5 A fuse | MR-S EHPS pump plug C pin 1 | 1055 | ST185-AB-engine | EPSEN |
| 84 | w_mrs_e | 20 | White | | EHPS pump speed | MR-S EHPS pump plug B pin 2 | 1055 | ST185-AB-engine | MRS |
| 85 | w_eps_rly_e | 18 | White | | EHPS relay coil (-) | MR-S EHPS pump plug B pin 5 | 1055 | ST185-AB-engine | EPS |
| 86 | | | | | spare, sealing plug | | | | |
| 87 | | | | | spare, sealing plug | | | | |
| 88 | | | | | spare, sealing plug | | | | |
| 89 | | | | | spare, sealing plug | | | | |
| 90 | | | | | spare, sealing plug | | | | |
| 91 | | | | | spare, sealing plug | | | | |
| 92 | | | | | spare, sealing plug | | | | |
| 93 | | | | | spare, sealing plug | | | | |
| 94 | | | | | spare, sealing plug | | | | |
| 95 | w_lam12_be14 | 12 | Red | | EFI relay k_efi -> LOGIC LAM 7.5 A fuse | CAN-Lambda +12 V splice - 12 AWG from BH-B pin 95 to the 16 AWG lead | 1157 | ST185-AB-engine | LAM |
| 96 | w_lamg_be14 | 12 | Black | | CAN-Lambda ground | CAN-Lambda ground splice - 12 AWG from BH-B pin 96 to the 16 AWG lead | 1157 | ST185-AB-engine | LAM |
| 97 | | | | | spare, sealing plug | | | | |
| 98 | | | | | spare, sealing plug | | | | |
| 99 | | | | | spare, sealing plug | | | | |
| 100 | | | | | spare, sealing plug | | | | |
| 101 | | | | | spare, sealing plug | | | | |
| 102 | | | | | spare, sealing plug | | | | |

**REVISIONS**

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| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 17 OF 52 | 432 x 395 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 4 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### BOOST-SOL - Boost Control Solenoid

housing 12052641 | contacts 12084200 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w219 | 20 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor | 1840 | ST185-AB-engine | SW12 |
| 2 | w103_e | 20 | White | ■■□■ | Boost solenoid (-) | BULKHEAD pin 58 | 2043 | ST185-AB-engine | BOOST |

### CAM - Cam Hall Sensor (RacerX)

housing TBD | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_rail_sp_8v_e | 20 | Red | □■□ | +8 V to cam sensor | BULKHEAD pin 63 | 1152 | ST185-AB-engine | 8V |
| 2 | cab_cam_c2 | 20 | Blue | ■■■□ | Cam sensor signal | BULKHEAD pin 64 | 1152 | ST185-AB-engine | CAM |
| 3 | w24 | 20 | Yellow | ■■■□ | Gnd cam sensor | BULKHEAD pin 65 | 1152 | ST185-AB-engine | CAM |

### CAN-LAMBDA - Link CAN-Lambda

housing DTM06-4S  + WM-4S | contacts 0462-005-2031 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_lam12_be | 16 | Red | ■□□ | EFI relay k_efi -> LOGIC LAM 7.5 A fuse | CAN-Lambda +12 V splice - 12 AWG from BH-B pin 95 to the 16 AWG lead | 200 | ST185-AB-engine | LAM |
| 2 | w_lamg_be | 16 | Black | ■□□ | CAN-Lambda ground | CAN-Lambda ground splice - 12 AWG from BH-B pin 96 to the 16 AWG lead | 200 | ST185-AB-engine | LAM |
| 3 | wc_l_lam_pin | 20 | Green | ■ | CAN L to CAN-Lambda | CAN L join - bulkhead, Lambda, 120 ohm end | 200 | ST185-AB-engine | CAN |
| 4 | wc_h_lam_pin | 20 | White | ■ | CAN H to CAN-Lambda | CAN H join - bulkhead, Lambda, 120 ohm end | 200 | ST185-AB-engine | CAN |

### CHARGE-IAT - Charge-Pipe IAT Sensor

housing 90980-11062 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w204_e | 20 | White | □■ | Charge temp signal | BULKHEAD pin 59 | 1893 | ST185-AB-engine | CPIAT |
| 2 | w10 | 20 | Blue | □■ | Gnd charge temp | Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice | 1690 | ST185-AB-engine | CPIAT |

### CLNT-P - Coolant Pressure Sensor

housing TBD | contacts TBD | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Coolant pressure pin table)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w36 | 20 | Blue | □■■ | Gnd coolant press | Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure | 810 | ST185-AB-engine | CLNTP |
| 2 | w_5v_brail_e_c47 | 20 | Orange | ■■ | +5 V coolant press | BULKHEAD pin 25 | 1013 | ST185-AB-engine | 5V |
| 3 | w35_e | 20 | White | □■■ | Coolant press signal | BULKHEAD pin 26 | 1013 | ST185-AB-engine | CLNTP |

### COIL-1 - COP 1 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w112 | 16 | Red | ■□■ | EFI relay k_efi -> LOGIC COP 15 A fuse | Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4 | 493 | ST185-AB-engine | COP |
| 2 | | | | | spare, not used | | | | |
| 3 | w120_e | 20 | White | □■■ | Coil 1 trigger (IGT) | BULKHEAD pin 9 | 1491 | ST185-AB-engine | COP1 |
| 4 | w_cop1g | 18 | Black | □■■ | Coil 1 ground | Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar) | 493 | ST185-AB-engine | COP1 |

### COIL-2 - COP 2 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w113 | 16 | Red | ■□■ | EFI relay k_efi -> LOGIC COP 15 A fuse | Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4 | 560 | ST185-AB-engine | COP |
| 2 | | | | | spare, not used | | | | |
| 3 | w121_e | 20 | White | □■□ | Coil 2 trigger (IGT) | BULKHEAD pin 10 | 1558 | ST185-AB-engine | COP2 |
| 4 | w_cop2g | 18 | Black | □■□ | Coil 2 ground | Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar) | 560 | ST185-AB-engine | COP2 |

### COIL-3 - COP 3 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w114 | 16 | Red | ■□■ | EFI relay k_efi -> LOGIC COP 15 A fuse | Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4 | 457 | ST185-AB-engine | COP |
| 2 | | | | | spare, not used | | | | |
| 3 | w122_e | 20 | White | □□■ | Coil 3 trigger (IGT) | BULKHEAD pin 11 | 1455 | ST185-AB-engine | COP3 |
| 4 | w_cop3g | 18 | Black | □□■ | Coil 3 ground | Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar) | 457 | ST185-AB-engine | COP3 |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 4 of 8)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 18 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 5 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### COIL-4 - COP 4 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w115 | 16 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4 | 352 | ST185-AB-engine | COP |
| 2 | | | | | spare, not used | | | | |
| 3 | w123_e | 20 | White | | Coil 4 trigger (IGT) | BULKHEAD pin 12 | 1350 | ST185-AB-engine | COP4 |
| 4 | w_cop4g | 18 | Black | | Coil 4 ground | Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar) | 352 | ST185-AB-engine | COP4 |

### CRANK - Crank VR Sensor (36-2)

housing DT06-2S-CE05  + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | cab_crank_c1 | 20 | White | | Crank sensor signal | BULKHEAD pin 74 | 1410 | ST185-AB-engine | CRANK |
| 2 | w2 | 20 | Blue | | Gnd crank sensor | BULKHEAD pin 75 | 1410 | ST185-AB-engine | CRANK |

### ECT Sensor

housing 90980-11062 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w201_e | 20 | White | | Coolant temp signal | BULKHEAD pin 70 | 2015 | ST185-AB-engine | ECT |
| 2 | w4 | 20 | Blue | | Gnd coolant temp | Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle | 1812 | ST185-AB-engine | ECT |

### ETB - Bosch 74.5mm ETB

housing D 261 205 358-01 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w215_e_ae | 18 | Black | | ETB motor - | BULKHEAD pin 46 | 1919 | ST185-AB-engine | ETBM |
| 2 | w127 | 20 | Yellow | | Gnd throttle sensor | Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle | 1716 | ST185-AB-engine | ETBS |
| 3 | w126 | 20 | Red | | +5 V throttle sensor | BULKHEAD pin 41 | 1919 | ST185-AB-engine | ETBS |
| 4 | w124_e_ae | 18 | Red | | ETB motor + | BULKHEAD pin 45 | 1919 | ST185-AB-engine | ETBM |
| 5 | w129_e | 20 | Blue | | Throttle pos 2 sub | BULKHEAD pin 44 | 1919 | ST185-AB-engine | ETBS |
| 6 | w128_e | 20 | Green | | Throttle pos 1 main | BULKHEAD pin 42 | 1919 | ST185-AB-engine | ETBS |

### FLEX - Flex Fuel Sensor

housing 13519047 | contacts 15326426 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Flex fuel pin table)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| A | w217 | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor | 1182 | ST185-AB-engine | SW12 |
| B | w47 | 20 | Yellow | | Gnd flex fuel | Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor | 315 | ST185-AB-engine | FLEX |
| C | w48_e | 20 | Blue | | Flex fuel signal | BULKHEAD pin 31 | 1385 | ST185-AB-engine | FLEX |

### FUEL-P - Fuel Pressure Sensor

housing 354404 | contacts in kit 354404 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Fuel pressure pin table)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w33 | 20 | Blue | | Gnd fuel pressure | Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed | 610 | ST185-AB-engine | FUELP |
| 2 | w_5v_brail_e_c46 | 20 | Orange | | +5 V fuel pressure | BULKHEAD pin 13 | 1504 | ST185-AB-engine | 5V |
| 3 | w32_e | 20 | White | | Fuel pressure signal | BULKHEAD pin 14 | 1504 | ST185-AB-engine | FUELP |

### IAT Sensor (Manifold)

housing 90980-11062 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w202_e | 20 | White | | Intake temp signal | BULKHEAD pin 33 | 1488 | ST185-AB-engine | IAT |
| 2 | w6 | 20 | Blue | | Gnd intake temp | Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle | 1285 | ST185-AB-engine | IAT |

### INJ-1 - Injector 1

housing 15419715 | contacts 12191819 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w211 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4 | 475 | ST185-AB-engine | INJ |
| 2 | w108_e | 20 | White | | Injector 1 drive (-) | BULKHEAD pin 21 | 1255 | ST185-AB-engine | INJ1 |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 5 of 8)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 19 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 6 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### INJ-2 - Injector 2

housing 15419715 | contacts 12191819 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w212 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4 | 388 | ST185-AB-engine | INJ |
| **2** | w109_e | 20 | White | | Injector 2 drive (-) | BULKHEAD pin 22 | 1168 | ST185-AB-engine | INJ2 |

### INJ-3 - Injector 3

housing 15419715 | contacts 12191819 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w213 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4 | 307 | ST185-AB-engine | INJ |
| **2** | w110_e | 20 | White | | Injector 3 drive (-) | BULKHEAD pin 23 | 1087 | ST185-AB-engine | INJ3 |

### INJ-4 - Injector 4

housing 15419715 | contacts 12191819 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w214 | 16 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4 | 242 | ST185-AB-engine | INJ |
| **2** | w111_e | 20 | White | | Injector 4 drive (-) | BULKHEAD pin 24 | 1022 | ST185-AB-engine | INJ4 |

### KNOCK-1 - Knock Sensor 1

housing 1 928 403 874 | contacts BDK 2.8 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_knock1_c1 | 20 | Blue | | Knock sensor signal | BULKHEAD pin 71 | 1051 | ST185-AB-engine | KNOCK1 |
| **2** | cab_knock1_c2 | 20 | White | | Knock sensor (-) | BULKHEAD pin 72 | 1051 | ST185-AB-engine | KNOCK1 |

### MAP Sensor

housing 354404 | contacts in kit 354404 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (MAP pin table)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w27 | 20 | Blue | | Gnd MAP sensor | Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor | 356 | ST185-AB-engine | MAP |
| **2** | w206 | 20 | Orange | | +5 V MAP sensor | BULKHEAD pin 16 | 1426 | ST185-AB-engine | MAP |
| **3** | w26_e | 20 | White | | MAP sensor signal | BULKHEAD pin 17 | 1426 | ST185-AB-engine | MAP |

### MRS-B - MR-S EHPS pump plug B

housing 90980-10897 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_mrs_e | 20 | White | | EHPS pump speed | BULKHEAD pin 84 | 1055 | ST185-AB-engine | MRS |
| **5** | w_eps_rly_e | 18 | White | | EHPS relay coil (-) | BULKHEAD pin 85 | 1055 | ST185-AB-engine | EPS |

### MRS-C - MR-S EHPS pump plug C

housing 90980-10942 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_hv_mrs_en_e | 20 | Red | | EFI relay k_efi -> LOGIC MRS 7.5 A fuse | BULKHEAD pin 83 | 1055 | ST185-AB-engine | EPSEN |

### OIL-P - Oil Pressure Sensor

housing 354404 | contacts in kit 354404 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Oil pressure pin table)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w30 | 20 | Blue | | Gnd oil pressure | Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure | 2045 | ST185-AB-engine | OILP |
| **2** | w_5v_brail_e_c45 | 20 | Orange | | +5 V oil pressure | BULKHEAD pin 27 | 2248 | ST185-AB-engine | 5V |
| **3** | w29_e | 20 | White | | Oil pressure signal | BULKHEAD pin 28 | 2248 | ST185-AB-engine | OILP |

### OIL-T - Oil Temp Sensor

housing 90980-11062 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w203_e | 20 | White | | Oil temp signal | BULKHEAD pin 61 | 1594 | ST185-AB-engine | OILT |
| **2** | w8 | 20 | Blue | | Gnd oil temp | Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice | 1391 | ST185-AB-engine | OILT |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 6 of 8)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 20 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 7 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### REV-SW - Back-up SW B1

housing 90980-11250 | contacts 8100-0461 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_sw7_e | 16 | Yellow | □ | Reverse sw +12 V out | BULKHEAD pin 78 | 2341 | ST185-AB-engine | REV |
| **2** | w_rev_feed_e | 16 | Red | □ | Back-up sw +12 V in | BULKHEAD pin 77 | 2341 | ST185-AB-engine | REV |

### TURBO-SPD - Turbo Speed Sensor

housing DT06-3S-CE05  + W3S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **A** | w218 | 20 | Red | yellow blue white | +5 V turbo speed | BULKHEAD pin 52 | 2167 | ST185-AB-engine | TURBO |
| **B** | w16_e | 20 | Blue | yellow blue white | Turbo speed signal | BULKHEAD pin 53 | 2167 | ST185-AB-engine | TURBO |
| **C** | w17 | 20 | Yellow | yellow blue white | Gnd turbo speed | BULKHEAD pin 56 | 2167 | ST185-AB-engine | TURBO |

### VSS - Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143

housing 90980-11143 | contacts 8100-0461 | pins per Toyota 3-wire speed sensor pinout (wilbo666 wiki; Daniel 2026-09-28)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_vss_12v | 20 | Red | blue | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | BULKHEAD pin 67 | 1578 | ST185-AB-engine | SW12 |
| **2** | w_vss_gnd | 20 | Yellow | yellow white yellow | Gnd gearbox speed | Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed | 684 | ST185-AB-engine | VSS |
| **3** | w_vss_sig_e | 20 | Blue | yellow white yellow | Gearbox speed signal | BULKHEAD pin 68 | 1578 | ST185-AB-engine | VSS |

### STARTER-TRIGGER - Starter solenoid S (50) terminal post, spade

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_strl_e | 12 | Red | white yellow | Starter solenoid S | BULKHEAD pin 3 | 2433 | ST185-AB-engine | STR |

### CAB-CAM cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_cam_sh | | Shield | blue blue yellow | Cam sensor shield | BULKHEAD pin 66 | 1152 | ST185-AB-engine | CAM |

### CAB-CAN-LAM cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_lamcan_be_sh | | Shield | yellow | CAN-Lambda shield | BULKHEAD pin 81 | 1157 | ST185-AB-engine | CAN |

### CAB-CRANK cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_crank_sh | | Shield | yellow blue | Crank sensor shield | BULKHEAD pin 76 | 1410 | ST185-AB-engine | CRANK |

### CAB-ETBM-AE cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_etbm_ae_sh | | Shield | blue yellow white | ETB motor shield | BULKHEAD pin 50 | 1919 | ST185-AB-engine | ETBM |

### CAB-ETBS cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_etbs_sh | | Shield | blue blue white | Throttle shield | BULKHEAD pin 47 | 1919 | ST185-AB-engine | ETBS |

### CAB-FLEX cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_flex_sh | | Shield | yellow yellow yellow | Flex fuel shield | BULKHEAD pin 32 | 1385 | ST185-AB-engine | FLEX |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 7 of 8)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 21 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 8 of 8)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### CAB-KNOCK1 cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_knock1_sh | | Shield | | Knock sensor shield | BULKHEAD pin 73 | 1051 | ST185-AB-engine | KNOCK1 |

### CAB-TURBO cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_turbo_sh | | Shield | | Turbo speed shield | BULKHEAD pin 57 | 2167 | ST185-AB-engine | TURBO |

### CAB-VSS cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_vss_sh | | Shield | | Gearbox speed shield | BULKHEAD pin 69 | 1578 | ST185-AB-engine | VSS |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS)  (part 8 of 8)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 22 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED  (part 1 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **27** | w_at_immob2 | 20 | White | | Anti-theft input | Alarm module - main harness (DT06-12SA) pin 9 | 380 | ST185-AntiTheft | IMMOB |

### SP-12V-SHK - Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_12v_shk | 20 | Red | | AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse | Shock / tilt sensor * pin A | 200 | ST185-AntiTheft | ALMS |

### SP-AT-12V - Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_12v | 20 | Red | | AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse | Alarm module - main harness (DT06-12SA) pin 1 | 200 | ST185-AntiTheft | ALM |

### SP-DOOR - Door trigger splice - door courtesy switches to alarm pin 6

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_door | 20 | Yellow | | Door trigger | Alarm module - main harness (DT06-12SA) pin 6 | 200 | ST185-AntiTheft | DOOR |
| | w_at_door_tie | 20 | Yellow | | Door trigger | OEM door courtesy switch line (R-W) at J/B No.1 connector 1B pin 11 (left kick panel); 0 V wit... | 1220 | ST185-AntiTheft | DOOR |

### SP-HOOD - Hood trigger splice - hood pin switch to alarm pin 7

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_hood | 20 | Yellow | | Hood trigger | Alarm module - main harness (DT06-12SA) pin 7 | 200 | ST185-AntiTheft | HOOD |
| | w_at_hood_tie | 20 | Yellow | | Hood trigger | Hood pin switch lead (new switch, RH front via loom C, spot TBD) | 2260 | ST185-AntiTheft | HOOD |

### SP-IG - Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_ig | 20 | Red | | EFI relay k_efi -> LOGIC ALMIG 2 A fuse | Alarm module - main harness (DT06-12SA) pin 3 | 200 | ST185-AntiTheft | ALMIG |

### SP-PARK - Park flash splice - alarm pin 5 to the park lamp relay

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_park | 20 | White | | Park lamp flash | Alarm module - main harness (DT06-12SA) pin 5 | 200 | ST185-AntiTheft | PARK |
| | w_at_park_tie | 20 | White | | Park lamp flash | Park lamp flash relay lead (cabin relay, part and spot TBD) | 1220 | ST185-AntiTheft | PARK |

### SP-TRUNK - Trunk trigger splice - trunk pin switch to alarm pin 8

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_trunk | 20 | Yellow | | Trunk trigger | Alarm module - main harness (DT06-12SA) pin 8 | 200 | ST185-AntiTheft | TRUNK |
| | w_at_trunk_tie | 20 | Yellow | | Trunk trigger | Trunk pin switch lead (new switch, spot TBD) | 4420 | ST185-AntiTheft | TRUNK |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED  (part 1 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| **ST185-harness-PIN** | **L** | **23 OF 52** | **432 x 279 mm** | **DG** | **2026-10-29** |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED  (part 2 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ALARM - Alarm module - main harness (DT06-12SA)

housing DT06-12SA  + W12S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_at_12v | 20 | Red | | AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse | Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1 | 200 | ST185-AntiTheft | ALM |
| **2** | w_at_gnd_a1 | 20 | Black | | Alarm ground | Cabin ground stud near the alarm (spot and EWD ID TBD, M6) | 200 | ST185-AntiTheft | ALM |
| **3** | w_at_ig | 20 | Red | | EFI relay k_efi -> LOGIC ALMIG 2 A fuse | Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3 | 200 | ST185-AntiTheft | ALMIG |
| **4** | w_at_siren | 20 | Red | | Siren + | Siren (engine bay, center firewall behind the engine) - leads cross via harness C's own RH fe... | 1660 | ST185-AntiTheft | SIREN |
| **5** | w_at_park | 20 | White | | Park lamp flash | Park flash splice - alarm pin 5 to the park lamp relay | 200 | ST185-AntiTheft | PARK |
| **6** | w_at_door | 20 | Yellow | | Door trigger | Door trigger splice - door courtesy switches to alarm pin 6 | 200 | ST185-AntiTheft | DOOR |
| **7** | w_at_hood | 20 | Yellow | | Hood trigger | Hood trigger splice - hood pin switch to alarm pin 7 | 200 | ST185-AntiTheft | HOOD |
| **8** | w_at_trunk | 20 | Yellow | | Trunk trigger | Trunk trigger splice - trunk pin switch to alarm pin 8 | 200 | ST185-AntiTheft | TRUNK |
| **9** | w_at_immob2 | 20 | White | | Anti-theft input | ECU-B (Link G4X plug B, centre dash) pin 27 | 380 | ST185-AntiTheft | IMMOB |
| **10** | w_at_led | 20 | Red | | Status LED + | Status LED (dash) pin 1 | 750 | ST185-AntiTheft | ALMLED |
| **11** | w_at_valet | 20 | Yellow | | Valet switch | Valet / program switch (hidden) pin 1 | 450 | ST185-AntiTheft | VALET |
| **12** | w_at_shk_sig | 20 | White | | Shock trigger | Shock / tilt sensor * pin C | 200 | ST185-AntiTheft | ALMS |

### SHOCK-SENSOR - Shock / tilt sensor *

housing DT06-3S  + W3S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **A** | w_at_12v_shk | 20 | Red | | AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse | Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A | 200 | ST185-AntiTheft | ALMS |
| **B** | w_at_gnd_b1 | 20 | Black | | Shock sensor ground | Cabin ground stud near the alarm (spot and EWD ID TBD, M6) | 200 | ST185-AntiTheft | ALMS |
| **C** | w_at_shk_sig | 20 | White | | Shock trigger | Alarm module - main harness (DT06-12SA) pin 12 | 200 | ST185-AntiTheft | ALMS |

### SIREN - Siren (engine bay, center firewall behind the engine) - leads cross via harness C's own RH fender grommet, decided 2026-10-10, not the FR wheel-speed grommet

housing DT06-2S-CE05  + W2S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_at_siren | 20 | Red | | Siren + | Alarm module - main harness (DT06-12SA) pin 4 | 1660 | ST185-AntiTheft | SIREN |
| **2** | w_at_gnd_a2 | 20 | Black | | Siren ground | Cabin ground stud near the alarm (spot and EWD ID TBD, M6) | 1660 | ST185-AntiTheft | SIREN |

### STATUS-LED - Status LED (dash)

housing DT06-2S  + W2S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_at_led | 20 | Red | | Status LED + | Alarm module - main harness (DT06-12SA) pin 10 | 750 | ST185-AntiTheft | ALMLED |
| **2** | w_at_gnd_b2 | 20 | Black | | Status LED gnd | Cabin ground stud near the alarm (spot and EWD ID TBD, M6) | 750 | ST185-AntiTheft | ALMLED |

### VALET-SW - Valet / program switch (hidden)

housing DT06-2S  + W2S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_at_valet | 20 | Yellow | | Valet switch | Alarm module - main harness (DT06-12SA) pin 11 | 450 | ST185-AntiTheft | VALET |
| **2** | w_at_gnd_c1 | 20 | Black | | Valet switch ground | Cabin ground stud near the alarm (spot and EWD ID TBD, M6) | 450 | ST185-AntiTheft | VALET |

### T-AT-DOOR - OEM door courtesy switch line (R-W) at J/B No.1 connector 1B pin 11 (left kick panel); 0 V with any door open

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_door_tie | 20 | Yellow | | Door trigger | Door trigger splice - door courtesy switches to alarm pin 6 | 1220 | ST185-AntiTheft | DOOR |

### T-AT-GND - Cabin ground stud near the alarm (spot and EWD ID TBD, M6)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_gnd_a1 | 20 | Black | | Alarm ground | Alarm module - main harness (DT06-12SA) pin 2 | 200 | ST185-AntiTheft | ALM |
| | w_at_gnd_a2 | 20 | Black | | Siren ground | Siren (engine bay, center firewall behind the engine) - leads cross via harness C's own RH fe... | 1660 | ST185-AntiTheft | SIREN |

### T-AT-GND-B - Cabin ground stud near the alarm (spot and EWD ID TBD, M6)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_gnd_b2 | 20 | Black | | Status LED gnd | Status LED (dash) pin 2 | 750 | ST185-AntiTheft | ALMLED |
| | w_at_gnd_b1 | 20 | Black | | Shock sensor ground | Shock / tilt sensor * pin B | 200 | ST185-AntiTheft | ALMS |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED  (part 2 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 24 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED  (part 3 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### T-AT-GND-C - Cabin ground stud near the alarm (spot and EWD ID TBD, M6)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_gnd_c1 | 20 | Black | | Valet switch ground | Valet / program switch (hidden) pin 2 | 450 | ST185-AntiTheft | VALET |

### T-AT-HOOD - Hood pin switch lead (new switch, RH front via loom C, spot TBD)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_hood_tie | 20 | Yellow | | Hood trigger | Hood trigger splice - hood pin switch to alarm pin 7 | 2260 | ST185-AntiTheft | HOOD |

### T-AT-PARK - Park lamp flash relay lead (cabin relay, part and spot TBD)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_park_tie | 20 | White | | Park lamp flash | Park flash splice - alarm pin 5 to the park lamp relay | 1220 | ST185-AntiTheft | PARK |

### T-AT-TRUNK - Trunk pin switch lead (new switch, spot TBD)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_trunk_tie | 20 | Yellow | | Trunk trigger | Trunk trigger splice - trunk pin switch to alarm pin 8 | 4420 | ST185-AntiTheft | TRUNK |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED  (part 3 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 25 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA  (part 1 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ECU-C - ECU Comms/Tuning Port

housing DTM06-6S  + WM-6S  |  contacts 0462-201-2031  |  pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_shd_can_trunk | 20 | Black | | CAN screen trunk | CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen | 650 | ST185-CAN | CAN |
| **2** | | | | | spare, sealing plug | | | | |
| **3** | wc_h_ecu | 20 | White | | CAN 1 high | CAN H join - ECU, CSB3, cluster, Pi, bulkhead | 250 | ST185-CAN | CAN |
| **4** | wc_l_ecu | 20 | Green | | CAN 1 low | CAN L join - ECU, CSB3, cluster, Pi, bulkhead | 250 | ST185-CAN | CAN |
| **5** | | | | | spare, sealing plug | | | | |
| **6** | | | | | spare, sealing plug | | | | |

### SP-CAN-SH-BH - CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_shd_can_bh | 20 | Black | | CAN screen trunk | CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen | 600 | ST185-CAN | CAN |

### SP-CAN-SH-CLU - CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_shd_can_trunk | 20 | Black | | CAN screen trunk | ECU Comms/Tuning Port pin 1 | 650 | ST185-CAN | CAN |
| | cab_can_ecu_sh | | Shield | | ECU CAN shield | cab_can_ecu screen (floats at the device end) | 650 | ST185-CAN | CAN |
| | cab_can_clu_sh | | Shield | | Cluster CAN shield | cab_can_clu screen (floats at the device end) | 200 | ST185-CAN | CAN |
| | w_shd_can_csb | 20 | Black | | CAN screen trunk | CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen | 1020 | ST185-CAN | CAN |
| | w_shd_can_bh | 20 | Black | | CAN screen trunk | CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda | 600 | ST185-CAN | CAN |

### SP-CAN-SH-CSB - CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_can_csb_sh | | Shield | | CSB3 CAN shield | cab_can_csb_csb screen (floats at the device end) | 200 | ST185-CAN | CAN |
| | w_shd_can_csb | 20 | Black | | CAN screen trunk | CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen | 1020 | ST185-CAN | CAN |
| | w_shd_can_rd | 20 | Black | | CAN screen trunk | CAN trunk, RealDash Pi branch - SP-CAN-SH-RD: CAN screen | 870 | ST185-CAN | CAN |

### SP-CAN-SH-RD - CAN trunk, RealDash Pi branch - SP-CAN-SH-RD: CAN screen

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_can_rd_sh | | Shield | | Pi CAN shield | cab_can_rd screen (floats at the device end) | 200 | ST185-CAN | CAN |
| | w_shd_can_rd | 20 | Black | | CAN screen trunk | CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen | 870 | ST185-CAN | CAN |

### SP-CANH - CAN H join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_h_ecu | 20 | White | | CAN 1 high | ECU Comms/Tuning Port pin 3 | 250 | ST185-CAN | CAN |
| | wc_term1 | 20 | Green | | ECU 120R CAN H | r_term_ecu pin 1 | 200 | ST185-CAN | CAN |
| | wc_h_csb | 20 | White | | CAN H to CSB3 | ECUMaster CSB3 - Power/CAN/Switch I/O pin 5 | 620 | ST185-CAN | CAN |
| | wc_h_clu | 20 | White | | CAN H to cluster | Cluster CAN board CANH + CANL pins (old meter spot) | 600 | ST185-CAN | CAN |
| | wc_h_rd | 20 | White | | CAN H to RealDash Pi | Pi USB-CAN adapter CANH + CANL pins (center dash) | 450 | ST185-CAN | CAN |

### SP-CANL - CAN L join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_l_ecu | 20 | Green | | CAN 1 low | ECU Comms/Tuning Port pin 4 | 250 | ST185-CAN | CAN |
| | wc_term2 | 20 | Green | | ECU 120R CAN L | r_term_ecu pin 2 | 200 | ST185-CAN | CAN |
| | wc_l_csb | 20 | Green | | CAN L to CSB3 | ECUMaster CSB3 - Power/CAN/Switch I/O pin 6 | 620 | ST185-CAN | CAN |
| | wc_l_clu | 20 | Green | | CAN L to cluster | Cluster CAN board CANH + CANL pins (old meter spot) | 600 | ST185-CAN | CAN |
| | wc_l_rd | 20 | Green | | CAN L to RealDash Pi | Pi USB-CAN adapter CANH + CANL pins (center dash) | 450 | ST185-CAN | CAN |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
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**ST185-harness - every wiring harness of the car, one drawing set**

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA  (part 1 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 26 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA  (part 2 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-DEV - Dash 12 V split - cluster + RealDash Pi, center dash

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_dev_clu | 16 | Red | | EFI relay k_efi -> LOGIC DEV 15 A fuse | Cluster buck converter IN+ pin (old meter spot) | 850 | ST185-CAN | DEV |
| | w_dev_rd | 16 | Red | | EFI relay k_efi -> LOGIC DEV 15 A fuse | Pi 12-to-5 V supply IN+ pin (center dash) | 200 | ST185-CAN | DEV |

### R-TERM-ECU - r_term_ecu

inline resistor 120R 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | wc_term1 | 20 | Green | | ECU 120R CAN H | CAN H join - ECU, CSB3, cluster, Pi, bulkhead | 200 | ST185-CAN | CAN |
| **2** | wc_term2 | 20 | Green | | ECU 120R CAN L | CAN L join - ECU, CSB3, cluster, Pi, bulkhead | 200 | ST185-CAN | CAN |

### CSB3 - ECUMaster CSB3 - Power/CAN/Switch I/O

housing HD36-24-33SE | contacts 0462-201-2031 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | wc_h_csb | 20 | White | | CAN H to CSB3 | CAN H join - ECU, CSB3, cluster, Pi, bulkhead | 620 | ST185-CAN | CAN |
| **6** | wc_l_csb | 20 | Green | | CAN L to CSB3 | CAN L join - ECU, CSB3, cluster, Pi, bulkhead | 620 | ST185-CAN | CAN |

### POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)

terminal 320575

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_clu | 16 | Black | | Cluster ground | Cluster buck converter IN- pin (old meter spot) | 1130 | ST185-CAN | DEV |
| | w_gnd_rd | 16 | Black | | RealDash Pi ground | Pi 12-to-5 V supply IN- pin (center dash) | 980 | ST185-CAN | DEV |

### T-CLUSTER-CAN - Cluster CAN board CANH + CANL pins (old meter spot)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_h_clu | 20 | White | | CAN H to cluster | CAN H join - ECU, CSB3, cluster, Pi, bulkhead | 600 | ST185-CAN | CAN |
| | wc_l_clu | 20 | Green | | CAN L to cluster | CAN L join - ECU, CSB3, cluster, Pi, bulkhead | 600 | ST185-CAN | CAN |

### T-CLUSTER-GND-LEAD - Cluster buck converter IN- pin (old meter spot)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_clu | 16 | Black | | Cluster ground | POWER box NEG bar M8 stud (glove box) | 1130 | ST185-CAN | DEV |

### T-CLUSTER-PWR - Cluster buck converter IN+ pin (old meter spot)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_dev_clu | 16 | Red | | EFI relay k_efi -> LOGIC DEV 15 A fuse | Dash 12 V split - cluster + RealDash Pi, center dash | 850 | ST185-CAN | DEV |

### T-REALDASH-CAN - Pi USB-CAN adapter CANH + CANL pins (center dash)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | wc_h_rd | 20 | White | | CAN H to RealDash Pi | CAN H join - ECU, CSB3, cluster, Pi, bulkhead | 450 | ST185-CAN | CAN |
| | wc_l_rd | 20 | Green | | CAN L to RealDash Pi | CAN L join - ECU, CSB3, cluster, Pi, bulkhead | 450 | ST185-CAN | CAN |

### T-REALDASH-GND-LEAD - Pi 12-to-5 V supply IN- pin (center dash)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_rd | 16 | Black | | RealDash Pi ground | POWER box NEG bar M8 stud (glove box) | 980 | ST185-CAN | DEV |

### T-REALDASH-PWR - Pi 12-to-5 V supply IN+ pin (center dash)

terminal flying lead (soldered, no terminal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_dev_rd | 16 | Red | | EFI relay k_efi -> LOGIC DEV 15 A fuse | Dash 12 V split - cluster + RealDash Pi, center dash | 200 | ST185-CAN | DEV |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA  (part 2 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 27 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA  (part 3 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### CAB-CAN-CLU cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_can_clu_sh | | Shield | | Cluster CAN shield | CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen | 200 | ST185-CAN | CAN |

### CAB-CAN-CSB-CSB cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_can_csb_sh | | Shield | | CSB3 CAN shield | CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen | 200 | ST185-CAN | CAN |

### CAB-CAN-ECU cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_can_ecu_sh | | Shield | | ECU CAN shield | CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen | 650 | ST185-CAN | CAN |

### CAB-CAN-RD cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_can_rd_sh | | Shield | | Pi CAN shield | CAN trunk, RealDash Pi branch - SP-CAN-SH-RD: CAN screen | 200 | ST185-CAN | CAN |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA  (part 3 of 3)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 28 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD  (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-CRUISE-LADDER - Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_an2 | 20 | Yellow | | Cruise ladder | OEM cruise stalk switch C14 pin 5, W wire pin 5 | 1220 | ST185-CSB3 | CRUISE |
| | w_cru_sig | 20 | Yellow | | Cruise ladder | r_cruise pin 2 | 200 | ST185-CSB3 | CRUISE |
| | w_cru_ladder_out | 20 | Yellow | | Cruise ladder | ECUMaster CSB3 - Power/CAN/Switch I/O pin 8 | 200 | ST185-CSB3 | CRUISE |

### SP-CSB-GND - CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_csbg | 20 | Black | | CSB3 ground | ECUMaster CSB3 - Power/CAN/Switch I/O pin 2 | 200 | ST185-CSB3 | CSB |
| | w_cru_gnd | 20 | Blue | | Cruise stalk ground | OEM cruise stalk switch C14 pin 17, BR wire pin 17 | 1220 | ST185-CSB3 | CSB |
| | w_opto_inc | 20 | Black | | Opto COM ground | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin COM1 | 200 | ST185-CSB3 | CSB |

### R-CRUISE - r_cruise

inline resistor 1k 1/4W

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_cru_sig | 20 | Yellow | | Cruise ladder | Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2 | 200 | ST185-CSB3 | CRUISE |

### CSB3 - ECUMaster CSB3 - Power/CAN/Switch I/O

housing HD36-24-33SE | contacts 0462-201-2031 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_csb12 | 20 | Red | | EFI relay k_efi -> LOGIC CSB 5 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 1050 | ST185-CSB3 | CSB |
| **2** | w_csbg | 20 | Black | | CSB3 ground | CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground | 200 | ST185-CSB3 | CSB |
| **3** | w_opto_vcc | 20 | Orange | | Opto +5 V | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin VCC | 200 | ST185-CSB3 | OPTO |
| **4** | w_opto_outc | 20 | Black | | Opto out ground | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin NEG | 200 | ST185-CSB3 | OPTO |
| **7** | | | | | spare, sealing plug | | | | |
| **8** | w_cru_ladder_out | 20 | Yellow | | Cruise ladder | Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2 | 200 | ST185-CSB3 | CRUISE |
| **9** | | | | | spare, sealing plug | | | | |
| **10** | | | | | spare, sealing plug | | | | |
| **11** | | | | | spare, sealing plug | | | | |
| **12** | | | | | spare, sealing plug | | | | |
| **13** | | | | | spare, sealing plug | | | | |
| **14** | | | | | spare, sealing plug | | | | |
| **15** | | | | | spare, sealing plug | | | | |
| **16** | | | | | spare, sealing plug | | | | |
| **17** | w_sw2 | 20 | Yellow | | Cruise MAIN on | OEM cruise stalk switch C14 pin 15, Y-B wire pin 15 | 1220 | ST185-CSB3 | CRUISE |
| **18** | | | | | spare, sealing plug | | | | |
| **19** | | | | | spare, sealing plug | | | | |
| **20** | w_sw5 | 20 | Yellow | | Clutch pedal sw | OEM cruise control clutch switch C17 pin 2, B wire pin 2 | 1460 | ST185-CSB3 | CLUTCH |
| **21** | w_opto_out1 | 20 | Yellow | | Brake to CSB3 S7 | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin OUT1 | 200 | ST185-CSB3 | BRAKE |
| **22** | w_opto_out2 | 20 | Yellow | | Reverse to CSB3 S8 | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin OUT2 | 200 | ST185-CSB3 | REV |
| **23** | | | | | spare, sealing plug | | | | |
| **24** | | | | | spare, sealing plug | | | | |
| **25** | | | | | spare, sealing plug | | | | |
| **27** | | | | | spare, sealing plug | | | | |
| **28** | | | | | spare, sealing plug | | | | |
| **29** | | | | | spare, sealing plug | | | | |
| **30** | | | | | spare, sealing plug | | | | |
| **31** | | | | | spare, sealing plug | | | | |
| **32** | | | | | spare, sealing plug | | | | |
| **33** | | | | | spare, sealing plug | | | | |

### LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968849-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **BUS2-5** | w_csb12 | 20 | Red | | EFI relay k_efi -> LOGIC CSB 5 A fuse | ECUMaster CSB3 - Power/CAN/Switch I/O pin 1 | 1050 | ST185-CSB3 | CSB |

### REVISIONS

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|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD  (part 1 of 2)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 29 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD  (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### OPTO-8 - Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3

housing TBD | contacts screw terminal (on the board) | pins per terminal names printed on the board

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **COM1** | w_opto_inc | 20 | Black | | Opto COM ground | CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground | 200 | ST185-CSB3 | CSB |
| **COM2** | | | | | spare, not used | | | | |
| **IN1** | w_sw6 | 16 | Yellow | | Brake on +12 V | OEM stop lamp switch S9 pin 1 (pedal), G-W wire pin 1 | 1460 | ST185-CSB3 | BRAKE |
| **IN3** | w_hl_sense_in | 20 | Red | | Lights on +12 V | OEM dimmer rheostat R10 pin 1 (left of the steering wheel), green TAIL pin 1 | 1220 | ST185-CSB3 | HLSENSE |
| **IN4** | | | | | spare, not used | | | | |
| **IN5** | | | | | spare, not used | | | | |
| **IN6** | | | | | spare, not used | | | | |
| **IN7** | | | | | spare, not used | | | | |
| **IN8** | | | | | spare, not used | | | | |
| **NEG** | w_opto_outc | 20 | Black | | Opto out ground | ECUMaster CSB3 - Power/CAN/Switch I/O pin 4 | 200 | ST185-CSB3 | OPTO |
| **OUT1** | w_opto_out1 | 20 | Yellow | | Brake to CSB3 S7 | ECUMaster CSB3 - Power/CAN/Switch I/O pin 21 | 200 | ST185-CSB3 | BRAKE |
| **OUT2** | w_opto_out2 | 20 | Yellow | | Reverse to CSB3 S8 | ECUMaster CSB3 - Power/CAN/Switch I/O pin 22 | 200 | ST185-CSB3 | REV |
| **OUT3** | w_hl_sense_out | 20 | Black | | Headlight sense to cluster | Centre cluster headlight-sense input (J8 pin 15, GPIO28, active-low) pin 15 | 1020 | ST185-CSB3 | HLSENSE |
| **OUT4** | | | | | spare, not used | | | | |
| **OUT5** | | | | | spare, not used | | | | |
| **OUT6** | | | | | spare, not used | | | | |
| **OUT7** | | | | | spare, not used | | | | |
| **OUT8** | | | | | spare, not used | | | | |
| **VCC** | w_opto_vcc | 20 | Orange | | Opto +5 V | ECUMaster CSB3 - Power/CAN/Switch I/O pin 3 | 200 | ST185-CSB3 | OPTO |

### T-CLUSTER-HL - Centre cluster headlight-sense input (J8 pin 15, GPIO28, active-low)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **15** | w_hl_sense_out | 20 | Black | | Headlight sense to cluster | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin OUT3 | 1020 | ST185-CSB3 | HLSENSE |

### OEM cruise control clutch switch C17 pin 2, B wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_sw5 | 20 | Yellow | | Clutch pedal sw | ECUMaster CSB3 - Power/CAN/Switch I/O pin 20 | 1460 | ST185-CSB3 | CLUTCH |

### OEM cruise stalk switch C14 pin 15, Y-B wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **15** | w_sw2 | 20 | Yellow | | Cruise MAIN on | ECUMaster CSB3 - Power/CAN/Switch I/O pin 17 | 1220 | ST185-CSB3 | CRUISE |

### OEM cruise stalk switch C14 pin 17, BR wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **17** | w_cru_gnd | 20 | Blue | | Cruise stalk ground | CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground | 1220 | ST185-CSB3 | CSB |

### OEM cruise stalk switch C14 pin 5, W wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | w_an2 | 20 | Yellow | | Cruise ladder | Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2 | 1220 | ST185-CSB3 | CRUISE |

### OEM dimmer rheostat R10 pin 1 (left of the steering wheel), green TAIL

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_hl_sense_in | 20 | Red | | Lights on +12 V | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin IN3 | 1220 | ST185-CSB3 | HLSENSE |

### OEM stop lamp switch S9 pin 1 (pedal), G-W wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_sw6 | 16 | Yellow | | Brake on +12 V | Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin IN1 | 1460 | ST185-CSB3 | BRAKE |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD  (part 2 of 2)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 30 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 1 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **20** | w81 | 20 | White | □□ | ETB relay coil (-) | B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)... | 680 | ST185-CabinPower | ETB |
| **28** | w76 | 18 | White | ■■ | EFI relay coil (-) | B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ... | 680 | ST185-CabinPower | EFI |

### ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-3 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | w82_ac | 18 | Red | □□ | ETB motor +12 V | B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)... | 680 | ST185-CabinPower | ETB |

### BUS-SW12 - SW12 bus plug DT06-12 (glove box, by LOGIC box)

housing DT06-12SA  + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_cp_sw12_in | 16 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | SW12 |
| **3** | w_k_fp_86_cp | 18 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ... | 2650 | ST185-CabinPower | SW12 |
| **4** | w_k_fan_86_cp | 18 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | B1-R4 Radiator fan relay (seated in the POWER box R4) pin 86 | 200 | ST185-CabinPower | SW12 |
| **5** | w_k_etb_86_cp | 20 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)... | 200 | ST185-CabinPower | SW12 |
| **7** | w_k_fan2_86_cp | 18 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | B1-R5 Condenser fan relay (seated in the POWER box R5) pin 86 | 200 | ST185-CabinPower | SW12 |
| **8** | w_k_eps_86_cp | 18 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro... | 200 | ST185-CabinPower | SW12 |
| **9** | w_k_eps_en_86_cp | 18 | Red | ■ | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85... | 200 | ST185-CabinPower | SW12 |
| **10** | | | | | spare, sealing plug | | | | |
| **11** | | | | | spare, sealing plug | | | | |
| **12** | | | | | spare, sealing plug | | | | |

### SP-12V-SHK - Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_12v_shk_tie_cp | 20 | Red | ■□□ | AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 860 | ST185-CabinPower | ALMS |

### SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fb_k_acc_30_cp | 16 | Red | ■■ | A/C clutch +12 V in | B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B... | 200 | ST185-CabinPower | ACC |
| | w_acc_fuse | 16 | Red | ■■ | ACC 10 A out | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | ACC |

### SP-AT-12V - Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_12v_tie_cp | 20 | Red | ■□■ | AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 860 | ST185-CabinPower | ALM |

### SP-DEV - Dash 12 V split - cluster + RealDash Pi, center dash

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_dev_cp | 16 | Red | ■■■ | EFI relay k_efi -> LOGIC DEV 15 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 980 | ST185-CabinPower | DEV |

### SP-HEAD-COIL - Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_hl_85_cp | 18 | White | ■■■ | Head LH coil (-) | B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 85 | 200 | ST185-CabinPower | HEAD |
| | w_hr_85_cp | 18 | White | ■■■ | Head RH coil (-) | B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 85 | 200 | ST185-CabinPower | HEAD |
| | w_head_coil_cp | 18 | White | ■■■ | Headlight coil (-) | OEM cowl plug IE1 pin 2 (LH kick), R-Y wire pin 2 | 1580 | ST185-CabinPower | HEAD |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 1 of 6)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 31 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 2 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-HL - HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_hl_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | HL |
| | w_hl_30_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse | B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 30 | 200 | ST185-CabinPower | HL |
| | w_hl_86_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse | B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 86 | 200 | ST185-CabinPower | HL |

### SP-HORN - Horn feed split - LH + RH horns, front centre

splice: crimp splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_pmu_horn_cp | 14 | Red | | LOGIC HZ 15 A fuse -> horn relay k_horn | B2-M3 Horn relay (seated in the LOGIC box M3) pin 87 | 2300 | ST185-CabinPower | HZ |

### SP-HR - HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_hr_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | HR |
| | w_hr_30_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse | B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 30 | 200 | ST185-CabinPower | HR |
| | w_hr_86_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse | B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 86 | 200 | ST185-CabinPower | HR |

### SP-HZ - HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_hz_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | HZ |
| | w_haz_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | OEM IE1 pin 12 (left kick, cowl side) pin 12 | 1580 | ST185-CabinPower | HZ |
| | w_horn_30_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | B2-M3 Horn relay (seated in the LOGIC box M3) pin 30 | 200 | ST185-CabinPower | HZ |
| | w_horn_86_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | B2-M3 Horn relay (seated in the LOGIC box M3) pin 86 | 200 | ST185-CabinPower | HZ |

### SP-IG - Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_at_ig_tie_cp | 20 | Red | | EFI relay k_efi -> LOGIC ALMIG 2 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 860 | ST185-CabinPower | ALMIG |

### DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018 + DRBF-1A | contacts 0460-220-1231 | pins per TE drawing DRB12-102PXE-L018 rev C3

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_cop_pwr_ac | 14 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 450 | ST185-CabinPower | COP |
| **2** | w_inj_pwr_ac | 14 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 450 | ST185-CabinPower | INJ |
| **3** | w_strl_c | 12 | Red | | Starter solenoid S | OEM R/B No.4 starter relay pin 4 (RH kick), starter out pin 4 | 700 | ST185-CabinPower | STR |
| **95** | w_lam12_cp | 14 | Red | | EFI relay k_efi -> LOGIC LAM 7.5 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 450 | ST185-CabinPower | LAM |

### K-ACC - B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B2-BUS2-8

housing 6-1419137-4 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_fb_k_acc_30_cp | 16 | Red | | A/C clutch +12 V in | SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86 | 200 | ST185-CabinPower | ACC |
| **87a** | | | | | spare, not used | | | | |

### K-EFI - B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) permanent 12 V (LOGIC EFI86 5 A)

housing 1-1414147-0 | contacts 280756-4 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w77_cp | 10 | Red | | EFI relay k_efi (LOGIC Bus 2 feed) | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | EFI |
| **85** | w76 | 18 | White | | EFI relay coil (-) | Link G4X plug A (centre dash) pin 28 | 680 | ST185-CabinPower | EFI |
| **86** | w_k_efi_86_cp | 18 | Red | | AMI BUS1 50 A fuse -> LOGIC EFI86 5 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | EFI |
| **87** | w_fb_k_efi_87_cp | 12 | Red | | AMI EFI 30 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 200 | ST185-CabinPower | EFI |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 2 of 6)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 32 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 3 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### K-EPS - B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to ground by the MR-S pump pin B5 (R22); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole plug (TE terminals 1/2, not marked 85/86), res...

housing V23132-A2001-B200 | contacts TBD | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_fb_k_eps_30_ac | 8 | Red | | AMI EPS 60 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 200 | ST185-CabinPower | EPS |
| **86** | w_k_eps_86_cp | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 8 | 200 | ST185-CabinPower | SW12 |
| **87** | w_eps_pwr_bc | 8 | Red | | AMI EPS 60 A fuse -> EHPS relay k_eps | MR-S EHPS pump plug A pin 1 | 1500 | ST185-CabinPower | EPS |
| **87a** | | | | | spare, not used | | | | |

### K-EPS-EN - B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V

housing 6-1419137-4 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_fb_k_eps_en_30_cp | 18 | Red | | EFI relay k_efi -> LOGIC EPSEN 7.5 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | EPSEN |
| **86** | w_k_eps_en_86_cp | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 9 | 200 | ST185-CabinPower | SW12 |
| **87** | w_k_eps_en_87_cp | 18 | Red | | LOGIC EPSEN 7.5 A fuse -> EPS enable relay k_eps_... | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | EPSEN |
| **87a** | | | | | spare, not used | | | | |

### K-ETB - B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V

housing 4-1904124-2 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_fb_k_etb_30_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC ETB 10 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | ETB |
| **85** | w81 | 20 | White | | ETB relay coil (-) | Link G4X plug A (centre dash) pin 20 | 680 | ST185-CabinPower | ETB |
| **86** | w_k_etb_86_cp | 20 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 5 | 200 | ST185-CabinPower | SW12 |
| **87** | w82_ac | 18 | Red | | ETB motor +12 V | ECU-B (Link G4X plug B, centre dash) pin 5 | 680 | ST185-CabinPower | ETB |
| **87a** | | | | | spare, not used | | | | |

### K-FAN - B1-R4 Radiator fan relay (seated in the POWER box R4)

housing 1-1414147-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **86** | w_k_fan_86_cp | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 4 | 200 | ST185-CabinPower | SW12 |
| **87** | w_fb_k_fan_87_cp | 8 | Red | | AMI FAN 40 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 200 | ST185-CabinPower | FAN |

### K-FAN2 - B1-R5 Condenser fan relay (seated in the POWER box R5)

housing 1-1393304-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **86** | w_k_fan2_86_cp | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 7 | 200 | ST185-CabinPower | SW12 |
| **87** | w_fb_k_fan2_87_ac | 8 | Red | | AMI FAN2 30 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 200 | ST185-CabinPower | FAN2 |

### K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts 282110-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **86** | w_k_fp_86_cp | 18 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 3 | 2650 | ST185-CabinPower | SW12 |

### K-HL - B2-M1 Headlight LH relay (seated in the LOGIC box M1)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_hl_30_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse | HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86 | 200 | ST185-CabinPower | HL |
| **85** | w_hl_85_cp | 18 | White | | Head LH coil (-) | Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2 | 200 | ST185-CabinPower | HEAD |
| **86** | w_hl_86_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse | HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86 | 200 | ST185-CabinPower | HL |
| **87a** | | | | | spare, not used | | | | |

### K-HORN - B2-M3 Horn relay (seated in the LOGIC box M3)

housing 4-1904124-3 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_horn_30_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay | 200 | ST185-CabinPower | HZ |
| **85** | w_horn_85_cp | 18 | White | | Horn relay coil (-) | OEM cowl plug IE1 pin 19 (LH kick), G wire pin 19 | 1580 | ST185-CabinPower | HZ |
| **86** | w_horn_86_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay | 200 | ST185-CabinPower | HZ |
| **87** | w_pmu_horn_cp | 14 | Red | | LOGIC HZ 15 A fuse -> horn relay k_horn | Horn feed split - LH + RH horns, front centre | 2300 | ST185-CabinPower | HZ |
| **87a** | | | | | spare, not used | | | | |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

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| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 33 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 4 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### K-HR - B2-M2 Headlight RH relay (seated in the LOGIC box M2)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_hr_30_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse | HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86 | 200 | ST185-CabinPower | HR |
| **85** | w_hr_85_cp | 18 | White | | Head RH coil (-) | Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2 | 200 | ST185-CabinPower | HEAD |
| **86** | w_hr_86_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse | HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86 | 200 | ST185-CabinPower | HR |
| **87a** | | | | | spare, not used | | | | |

### LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 2R5 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **BUS1** | w_hc_fb | 8 | Red | | AMI BUS1 50 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 200 | ST185-CabinPower | BUS1 |
| **BUS1-1** | w_k_efi_86_cp | 18 | Red | | AMI BUS1 50 A fuse -> LOGIC EFI86 5 A fuse | B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ... | 200 | ST185-CabinPower | EFI |
| **BUS1-10** | w_fb_k_etb_30_cp | 16 | Red | | AMI BUS1 50 A fuse -> LOGIC ETB 10 A fuse | B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)... | 200 | ST185-CabinPower | ETB |
| **BUS1-2** | w_hl_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse | HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86 | 200 | ST185-CabinPower | HL |
| **BUS1-3** | w_hr_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse | HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86 | 200 | ST185-CabinPower | HR |
| **BUS1-4** | w_hz_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay | 200 | ST185-CabinPower | HZ |
| **BUS1-7** | w_at_12v_tie_cp | 20 | Red | | AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse | Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1 | 860 | ST185-CabinPower | ALM |
| **BUS1-8** | w_at_12v_shk_tie_cp | 20 | Red | | AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse | Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A | 860 | ST185-CabinPower | ALMS |
| **BUS2** | w77_cp | 10 | Red | | EFI relay k_efi (LOGIC Bus 2 feed) | B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ... | 200 | ST185-CabinPower | EFI |
| **BUS2-1** | w_lam12_cp | 14 | Red | | EFI relay k_efi -> LOGIC LAM 7.5 A fuse | DRB102 receptacle (RH firewall) pin 95 | 450 | ST185-CabinPower | LAM |
| **BUS2-12 IN** | w_k_eps_en_87_cp | 18 | Red | | LOGIC EPSEN 7.5 A fuse -> EPS enable relay k_eps_... | B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85... | 200 | ST185-CabinPower | EPSEN |
| **BUS2-13 IN** | w_start_in_cp | 20 | Yellow | | Start sw ST1 feed | OEM ST1 lead B-W at old R/B No.4 pin 1, RH kick panel pin 1 | 450 | ST185-CabinPower | B30 |
| **BUS2-2** | w_cp_sw12_in | 16 | Red | | EFI relay k_efi -> LOGIC SW12 7.5 A fuse | SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 1 | 200 | ST185-CabinPower | SW12 |
| **BUS2-3** | w_cop_pwr_ac | 14 | Red | | EFI relay k_efi -> LOGIC COP 15 A fuse | DRB102 receptacle (RH firewall) pin 1 | 450 | ST185-CabinPower | COP |
| **BUS2-4** | w_inj_pwr_ac | 14 | Red | | EFI relay k_efi -> LOGIC INJ 15 A fuse | DRB102 receptacle (RH firewall) pin 2 | 450 | ST185-CabinPower | INJ |
| **BUS2-6** | w_dev_cp | 16 | Red | | EFI relay k_efi -> LOGIC DEV 15 A fuse | Dash 12 V split - cluster + RealDash Pi, center dash | 980 | ST185-CabinPower | DEV |
| **BUS2-7** | w_fb_k_eps_en_30_cp | 18 | Red | | EFI relay k_efi -> LOGIC EPSEN 7.5 A fuse | B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85... | 200 | ST185-CabinPower | EPSEN |
| **BUS2-8** | w_acc_fuse | 16 | Red | | EFI relay k_efi -> LOGIC ACC 10 A fuse | SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86 | 200 | ST185-CabinPower | ACC |
| **BUS2-9** | w_at_ig_tie_cp | 20 | Red | | EFI relay k_efi -> LOGIC ALMIG 2 A fuse | Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3 | 860 | ST185-CabinPower | ALMIG |

### MRS-A - MR-S EHPS pump plug A

housing 90980-12068 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_eps_pwr_bc | 8 | Red | | AMI EPS 60 A fuse -> EHPS relay k_eps | B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro... | 1500 | ST185-CabinPower | EPS |

### POWER - B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fusebox rev 7, DECISIONS R75, R91): the tinned copper + bar IS the AMI bar (7 AMI IN relay feeds, 7 AMI OUT; no 1/0 link cable), NEG ...

housing ST185-FB-POWER rev 7 | contacts 2R5 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **AMI IN-1** | w_rb4_feed | 8 | Red | | AMI HTR 40 A fuse | OEM R/B No.4 40 A HEATER fuse pin 1 (RH kick) pin 1 | 450 | ST185-CabinPower | HTR |
| **AMI IN-2** | w_fb_k_efi_87_cp | 12 | Red | | AMI EFI 30 A fuse | B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ... | 200 | ST185-CabinPower | EFI |
| **AMI IN-4** | w_fb_k_fan_87_cp | 8 | Red | | AMI FAN 40 A fuse | B1-R4 Radiator fan relay (seated in the POWER box R4) pin 87 | 200 | ST185-CabinPower | FAN |
| **AMI IN-5** | w_fb_k_fan2_87_ac | 8 | Red | | AMI FAN2 30 A fuse | B1-R5 Condenser fan relay (seated in the POWER box R5) pin 87 | 200 | ST185-CabinPower | FAN2 |
| **AMI IN-6** | w_rb2_feed | 10 | Red | | AMI PWR 30 A fuse | OEM R/B No.2 30 A POWER fuse pin 1 (LH kick) pin 1 | 1580 | ST185-CabinPower | PWR |
| **AMI OUT-1** | w_hc_fb | 8 | Red | | AMI BUS1 50 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 200 | ST185-CabinPower | BUS1 |
| **AMI OUT-5** | w_pdb_jb1 | 6 | Red | | AMI JB1 60 A fuse | OEM J/B No.1 1l pin 1 (left kick) pin 1 | 1580 | ST185-CabinPower | JB1 |
| **AMI OUT-6** | w_fb_k_eps_30_ac | 8 | Red | | AMI EPS 60 A fuse | B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro... | 200 | ST185-CabinPower | EPS |

### FOG-CAP-1 - Cut and capped fog lead (circuit removed)

terminal heat-shrink cap

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fog_rb3_1 | 18 | White | | Fog circuit removed | OEM R/B No.3 fog relay pin 1 (fog lamp feed), RH kick / dash pin 1 | 200 | ST185-CabinPower | FOG |

### FOG-CAP-4 - Cut and capped fog lead (circuit removed)

terminal heat-shrink cap

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fog_rb3_4 | 18 | White | | Fog circuit removed | OEM R/B No.3 fog relay pin 4 (20 A FR FOG in), RH kick / dash pin 4 | 200 | ST185-CabinPower | FOG |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 4 of 6)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 34 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 5 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### FOG-CAP-6 - Cut and capped fog lead (circuit removed)

terminal heat-shrink cap

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fog_sw | 18 | White | | Fog circuit removed | OEM fog light switch F12 pin 6 (coil-), RH kick / dash pin 6 | 1580 | ST185-CabinPower | FOG |

### OEM fog light switch F12 pin 6 (coil-), RH kick / dash

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **6** | w_fog_sw | 18 | White | | Fog circuit removed | Cut and capped fog lead (circuit removed) | 1580 | ST185-CabinPower | FOG |

### OEM IE1 pin 12 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **12** | w_haz_cp | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse | HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay | 1580 | ST185-CabinPower | HZ |

### OEM cowl plug IE1 pin 19 (LH kick), G wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **19** | w_horn_85_cp | 18 | White | | Horn relay coil (-) | B2-M3 Horn relay (seated in the LOGIC box M3) pin 85 | 1580 | ST185-CabinPower | HZ |

### OEM cowl plug IE1 pin 2 (LH kick), R-Y wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_head_coil_cp | 18 | White | | Headlight coil (-) | Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2 | 1580 | ST185-CabinPower | HEAD |

### OEM J/B No.1 1I pin 1 (left kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_pdb_jb1 | 6 | Red | | AMI JB1 60 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 1580 | ST185-CabinPower | JB1 |

### OEM R/B No.2 30 A POWER fuse pin 1 (LH kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_rb2_feed | 10 | Red | | POWER fuse feed | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 1580 | ST185-CabinPower | PWR |

### OEM R/B No.3 fog relay pin 1 (fog lamp feed), RH kick / dash

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_fog_rb3_1 | 18 | White | | Fog circuit removed | Cut and capped fog lead (circuit removed) | 200 | ST185-CabinPower | FOG |

### OEM R/B No.3 fog relay pin 4 (20 A FR FOG in), RH kick / dash

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_fog_rb3_4 | 18 | White | | Fog circuit removed | Cut and capped fog lead (circuit removed) | 200 | ST185-CabinPower | FOG |

### OEM R/B No.4 40 A HEATER fuse pin 1 (RH kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_rb4_feed | 8 | Red | | Heater fuse feed | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 450 | ST185-CabinPower | HTR |

### OEM ST1 lead B-W at old R/B No.4 pin 1, RH kick panel

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_start_in_cp | 20 | Yellow | | Start sw ST1 feed | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 450 | ST185-CabinPower | B30 |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 5 of 6)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 35 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 6 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

**OEM R/B No.4 starter relay pin 4 (RH kick), starter out**

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_strl_c | 12 | Red | □■ | Starter solenoid S | DRB102 receptacle (RH firewall) pin 3 | 700 | ST185-CabinPower | STR |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP  (part 6 of 6)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 36 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12  (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-LED-12V - LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_ig_tap | 20 | Red | | LED IG +12 V | OEM cluster plug C12 pin 9, B-O wire pin 9 | 200 | ST185-ClusterLED | LIG |
| | w_chg_a | 20 | Red | | Charge LED + | Charge LED pin 1 | 200 | ST185-ClusterLED | LCHG |
| | w_oil_a | 20 | Red | | Oil LED + | Oil pressure LED pin 1 | 200 | ST185-ClusterLED | LOIL |
| | w_pk_a | 20 | Red | | Park brake LED + | Park brake LED pin 1 | 200 | ST185-ClusterLED | LPK |

### SP-LED-GND - LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_hb_k | 20 | Black | | High beam LED gnd | High beam LED pin 2 | 200 | ST185-ClusterLED | LHB |
| | w_tl_k | 20 | Black | | Turn LH LED gnd | Turn LH LED pin 2 | 200 | ST185-ClusterLED | LTL |
| | w_tr_k | 20 | Black | | Turn RH LED gnd | Turn RH LED pin 2 | 200 | ST185-ClusterLED | LTR |
| | w_led_gnd_out | 20 | Black | | LED common ground | Ground stud, LH kick panel (EWD ground point ID) | 1550 | ST185-ClusterLED | LGND |

### D-CHG - Charge LED

inline diode TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_chg_a | 20 | Red | | Charge LED + | LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs | 200 | ST185-ClusterLED | LCHG |
| **2** | w_chg_k | 20 | White | | Charge lamp drive | OEM cluster plug C12 pin 8, Y wire pin 8 | 200 | ST185-ClusterLED | LCHG |

### D-HB - High beam LED

inline diode TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_hb_a | 20 | Red | | High beam LED + | OEM cluster plug C11 pin 12, R-L wire pin 12 | 200 | ST185-ClusterLED | LHB |
| **2** | w_hb_k | 20 | Black | | High beam LED gnd | LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud | 200 | ST185-ClusterLED | LHB |

### D-OIL - Oil pressure LED

inline diode TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_oil_a | 20 | Red | | Oil LED + | LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs | 200 | ST185-ClusterLED | LOIL |
| **2** | w_oil_k | 20 | White | | Oil LED sink (CSB3) | ECUMaster CSB3 - Power/CAN/Switch I/O pin 26 | 1020 | ST185-ClusterLED | LOIL |

### D-PK - Park brake LED

inline diode TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_pk_a | 20 | Red | | Park brake LED + | LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs | 200 | ST185-ClusterLED | LPK |
| **2** | w_pk_k | 20 | Yellow | | Park brake sw (-) | OEM cluster plug C12 pin 1, R-G wire pin 1 | 200 | ST185-ClusterLED | LPK |

### D-TL - Turn LH LED

inline diode TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_tl_a | 20 | Red | | Turn LH LED + | OEM cluster plug C11 pin 2, G-B wire pin 2 | 200 | ST185-ClusterLED | LTL |
| **2** | w_tl_k | 20 | Black | | Turn LH LED gnd | LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud | 200 | ST185-ClusterLED | LTL |

### D-TR - Turn RH LED

inline diode TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_tr_a | 20 | Red | | Turn RH LED + | OEM cluster plug C11 pin 11, G-Y wire pin 11 | 200 | ST185-ClusterLED | LTR |
| **2** | w_tr_k | 20 | Black | | Turn RH LED gnd | LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud | 200 | ST185-ClusterLED | LTR |

### CSB3 - ECUMaster CSB3 - Power/CAN/Switch I/O

housing HD36-24-33SE | contacts 0462-201-2031 | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **26** | w_oil_k | 20 | White | | Oil LED sink (CSB3) | Oil pressure LED pin 2 | 1020 | ST185-ClusterLED | LOIL |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12  (part 1 of 2)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 37 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12  (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### GND-ID - Ground stud, LH kick panel (EWD ground point ID)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_led_gnd_out | 20 | Black | | LED common ground | LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud | 1550 | ST185-ClusterLED | LGND |

### OEM cluster plug C11 pin 11, G-Y wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **11** | w_tr_a | 20 | Red | | Turn RH LED + | Turn RH LED pin 1 | 200 | ST185-ClusterLED | LTR |

### OEM cluster plug C11 pin 12, R-L wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **12** | w_hb_a | 20 | Red | | High beam LED + | High beam LED pin 1 | 200 | ST185-ClusterLED | LHB |

### OEM cluster plug C11 pin 2, G-B wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_tl_a | 20 | Red | | Turn LH LED + | Turn LH LED pin 1 | 200 | ST185-ClusterLED | LTL |

### OEM cluster plug C12 pin 1, R-G wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_pk_k | 20 | Yellow | | Park brake sw (-) | Park brake LED pin 2 | 200 | ST185-ClusterLED | LPK |

### OEM cluster plug C12 pin 8, Y wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **8** | w_chg_k | 20 | White | | Charge lamp drive | Charge LED pin 2 | 200 | ST185-ClusterLED | LCHG |

### OEM cluster plug C12 pin 9, B-O wire

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **9** | w_ig_tap | 20 | Red | | LED IG +12 V | LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs | 200 | ST185-ClusterLED | LIG |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12  (part 2 of 2)

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 38 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 1 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-CLR - Clearance feed split - F7 LH + F8 RH

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_clr | 16 | Red | | Clearance lamp feed | OEM J/B No.1 1H pin 6 (left kick) pin 1H-6 | 3680 | ST185-EngineRoom-C | CLR |
| | w_clr_f7 | 16 | Red | | LH clearance + | LH front turn / clearance lamp F7 pin 1 pin 1 | 900 | ST185-EngineRoom-C | CLR |
| | w_clr_f8 | 16 | Red | | RH clearance + | RH front turn / clearance lamp F8 pin 1 pin 1 | 900 | ST185-EngineRoom-C | CLR |

### SP-HL-HIGH - High-beam return join - H1 + H2 pin 1 to IE1 pin 11

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_hl_high_h1 | 16 | Red | | LH high-beam return | LH headlamp H1 pin 1 pin 1 | 900 | ST185-EngineRoom-C | HHI |
| | w_hl_high_h2 | 16 | Red | | RH high-beam return | RH headlamp H2 pin 1 pin 1 | 900 | ST185-EngineRoom-C | HHI |
| | w_hl_high | 14 | Red | | High-beam return | OEM IE1 pin 11 (left kick, cowl side) pin 11 | 3680 | ST185-EngineRoom-C | HHI |

### SP-HL-LOW - Low-beam return join - H1 + H2 pin 2 to IE1 pin 18

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_hl_low_h1 | 16 | Red | | LH low-beam return | LH headlamp H1 pin 2 pin 2 | 900 | ST185-EngineRoom-C | HLO |
| | w_hl_low_h2 | 16 | Red | | RH low-beam return | RH headlamp H2 pin 2 pin 2 | 900 | ST185-EngineRoom-C | HLO |
| | w_hl_low | 14 | Red | | Low-beam return | OEM IE1 pin 18 (left kick, cowl side) pin 18 | 3680 | ST185-EngineRoom-C | HLO |

### SP-HORN - Horn feed split - LH + RH horns, front centre

splice: crimp splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_horn_lh | 16 | Red | | LOGIC HZ 15 A fuse -> horn relay k_horn | Horn LH (EWD H3) - custom DT 2-way pin 1 | 200 | ST185-EngineRoom-C | HZ |
| | w_horn_rh | 16 | Red | | LOGIC HZ 15 A fuse -> horn relay k_horn | Horn RH (EWD H4) - custom DT 2-way pin 1 | 200 | ST185-EngineRoom-C | HZ |

### SP-RTR-CLOSE - Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rtr_close | 18 | Yellow | | Retract close limit | OEM IE1 pin 9 (left kick, cowl side) pin 9 | 3680 | ST185-EngineRoom-C | RTM |
| | w_rtr_close_r2 | 18 | Yellow | | Retract close limit | LH retract motor R2 pin 3 pin 3 | 900 | ST185-EngineRoom-C | RTM |
| | w_rtr_close_r3 | 18 | Yellow | | Retract close limit | RH retract motor R3 pin 3 pin 3 | 900 | ST185-EngineRoom-C | RTM |

### SP-RTR-GND-R2 - LH retract motor ground - 16 AWG pigtail to 12 AWG, at R2

splice: crimp splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rtr_gnd_r2 | 16 | Black | | LH retract ground | LH retract motor R2 pin 5 pin 5 | 200 | ST185-EngineRoom-C | RTM |
| | w_rtr_gnd_r2_t | 12 | Black | | LH retract ground | Ground stud EB, LH front fender (EWD ground point EB) | 200 | ST185-EngineRoom-C | RTM |

### SP-RTR-GND-R3 - RH retract motor ground - 16 AWG pigtail to 12 AWG, at R3

splice: crimp splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rtr_gnd_r3 | 16 | Black | | RH retract ground | RH retract motor R3 pin 5 pin 5 | 200 | ST185-EngineRoom-C | RTM |
| | w_rtr_gnd_r3_t | 12 | Black | | RH retract ground | Ground stud EB, LH front fender (EWD ground point EB) | 1600 | ST185-EngineRoom-C | RTM |

### SP-RTR-OPEN - Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rtr_open | 18 | Yellow | | Retract open limit | OEM IE1 pin 16 (left kick, cowl side) pin 16 | 3680 | ST185-EngineRoom-C | RTM |
| | w_rtr_open_r2 | 18 | Yellow | | Retract open limit | LH retract motor R2 pin 1 pin 1 | 900 | ST185-EngineRoom-C | RTM |
| | w_rtr_open_r3 | 18 | Yellow | | Retract open limit | RH retract motor R3 pin 1 pin 1 | 900 | ST185-EngineRoom-C | RTM |

### SP-RTR-R2-4 - LH retract motor feed - 12 AWG to 16 AWG pigtail, at R2

splice: crimp splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rtr_r2_4 | 12 | Red | | LH retract motor 4 | OEM IE1 pin 14 (left kick, cowl side) pin 14 | 4380 | ST185-EngineRoom-C | RTM |
| | w_rtr_r2_4_p | 16 | Red | | LH retract motor 4 | LH retract motor R2 pin 4 pin 4 | 200 | ST185-EngineRoom-C | RTM |

**REVISIONS**

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|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 1 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 39 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 2 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-RTR-R3-4 - RH retract motor feed - 12 AWG to 16 AWG pigtail, at R3

splice: crimp splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rtr_r3_4 | 12 | Red | | RH retract motor 4 | OEM IE1 pin 7 (left kick, cowl side) pin 7 | 2980 | ST185-EngineRoom-C | RTM |
| | w_rtr_r3_4_p | 16 | Red | | RH retract motor 4 | RH retract motor R3 pin 4 pin 4 | 200 | ST185-EngineRoom-C | RTM |

### ALT-PLUG - Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout)

housing 90980-10341 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_alt_s_c | 20 | White | | Alt S battery sense | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 2660 | ST185-EngineRoom-C | ALTS |
| 2 | w_alt_l_c | 20 | White | | Alt L charge lamp | OEM IG2 pin 16 (under the ECU, cowl side) pin 16 | 3140 | ST185-EngineRoom-C | ALTL |

### FAN-COND - Uprated condenser fan (+)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_fan2_pwr | 8 | Red | | Cond fan +12 V | B1-R5 Condenser fan relay (seated in the POWER box R5) pin 30 | 1900 | ST185-EngineRoom-C | FAN2 |

### FAN-COND-GND - Uprated condenser fan (GND)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_fan2_gnd | 8 | Black | | Cond fan ground | Ground stud EA, RH front fender (EWD ground point EA) | 500 | ST185-EngineRoom-C | FAN2 |

### FAN-RAD - Uprated radiator fan (+)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_fan_pwr | 8 | Red | | AMI FAN 40 A fuse -> rad fan relay k_fan | B1-R4 Radiator fan relay (seated in the POWER box R4) pin 30 | 2200 | ST185-EngineRoom-C | FAN |

### FAN-RAD-GND - Uprated radiator fan (GND)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_fan_gnd | 8 | Black | | Rad fan ground | Ground stud EA, RH front fender (EWD ground point EA) | 800 | ST185-EngineRoom-C | FAN |

### HORN-LH - Horn LH (EWD H3) - custom DT 2-way

housing DT06-2S-CE05  + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_horn_lh | 16 | Red | | LOGIC HZ 15 A fuse -> horn relay k_horn | Horn feed split - LH + RH horns, front centre | 200 | ST185-EngineRoom-C | HZ |
| 2 | w_horn_lh_g | 16 | Black | | Horn LH ground | Ground stud EA, RH front fender (EWD ground point EA) | 900 | ST185-EngineRoom-C | HZ |

### HORN-RH - Horn RH (EWD H4) - custom DT 2-way

housing DT06-2S-CE05  + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_horn_rh | 16 | Red | | LOGIC HZ 15 A fuse -> horn relay k_horn | Horn feed split - LH + RH horns, front centre | 200 | ST185-EngineRoom-C | HZ |
| 2 | w_horn_rh_g | 16 | Black | | Horn RH ground | Ground stud EA, RH front fender (EWD ground point EA) | 900 | ST185-EngineRoom-C | HZ |

### K-ACC - B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B2-BUS2-8

housing 6-1419137-4 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 87 | w_acc_clutch | 16 | Red | | A/C clutch +12 V | A/C compressor A4 pin 4 pin 4 | 1350 | ST185-EngineRoom-C | ACC |

### K-FAN - B1-R4 Radiator fan relay (seated in the POWER box R4)

housing 1-1414147-0 | contacts 280755-4 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 30 | w_fan_pwr | 8 | Red | | AMI FAN 40 A fuse -> rad fan relay k_fan | Uprated radiator fan (+) pin 1 | 2200 | ST185-EngineRoom-C | FAN |

**REVISIONS**

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| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 2 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 40 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 3 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### K-FAN2 - B1-R5 Condenser fan relay (seated in the POWER box R5)

housing 1-1393304-0 | contacts 280755-4 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_fan2_pwr | 8 | Red | □■ | Cond fan +12 V | Uprated condenser fan (+) pin 1 | 1900 | ST185-EngineRoom-C | FAN2 |

### K-HL - B2-M1 Headlight LH relay (seated in the LOGIC box M1)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **87** | w_pmu_hl | 14 | Red | ■■■ | LOGIC HL 15 A fuse -> HEAD LH relay k_hl | LH headlamp H1 pin 3 pin 3 | 3000 | ST185-EngineRoom-C | HL |

### K-HR - B2-M2 Headlight RH relay (seated in the LOGIC box M2)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **87** | w_pmu_hr | 14 | Red | ■■□ | LOGIC HR 15 A fuse -> HEAD RH relay k_hr | RH headlamp H2 pin 3 pin 3 | 1600 | ST185-EngineRoom-C | HR |

### LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968851-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **BUS1-5** | w_pmu_dome | 14 | Red | ■□ | AMI BUS1 50 A fuse -> LOGIC DM 20 A fuse | OEM J/B No.1 1H pin 2 (left kick) pin 2 | 1580 | ST185-EngineRoom-C | DM |
| **BUS1-6** | w_rb2_wl | 14 | Red | ■■ | AMI BUS1 50 A fuse -> LOGIC WL 15 A fuse | OEM W-L lead at old R/B2 pin 2 (left kick) pin 2 | 1580 | ST185-EngineRoom-C | WL |
| **BUS1-9** | w_alt_s_c | 20 | White | □■ | Alt S battery sense | Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout) pin 1 | 2660 | ST185-EngineRoom-C | ALTS |

### MRS-A - MR-S EHPS pump plug A

housing 90980-12068 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_eps_gnd | 8 | Black | ■■■ | EHPS pump ground | Engine block ground stud, lug 3 (Route 1 point 5) | 1750 | ST185-EngineRoom-C | EPS |

### MRS-B - MR-S EHPS pump plug B

housing 90980-10897 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | | | | | spare, not used | | | | |
| **3** | | | | | spare, not used | | | | |
| **4** | | | | | spare, not used | | | | |
| **6** | | | | | spare, not used | | | | |

### MRS-C - MR-S EHPS pump plug C

housing 90980-10942 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | | | | | spare, not used | | | | |

### POWER - B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fusebox rev 7, DECISIONS R75, R91): the tinned copper + bar IS the AMI bar (7 AMI IN relay feeds, 7 AMI OUT; no 1/0 link cable), NEG ...

housing ST185-FB-POWER rev 7 | contacts 2R5 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **AMI OUT-2** | w_pmu_rtr | 10 | Red | ■■■ | AMI RTR 30 A fuse | OEM IE1 pin 3 (left kick, cowl side) pin 3 | 1580 | ST185-EngineRoom-C | RTR |
| **AMI OUT-3** | w_pdb_am1 | 8 | Red | ■■□ | AMI AM1 40 A fuse | OEM IE1 pin 10 (left kick, cowl side) pin 10 | 1580 | ST185-EngineRoom-C | AM1 |
| **AMI OUT-4** | w_pdb_am2 | 10 | Red | ■□■ | AMI AM2 30 A fuse | OEM IE1 pin 17 (left kick, cowl side) pin 17 | 1580 | ST185-EngineRoom-C | AM2 |

### POWER-STUDS - POWER box bus zone (rev 7, glove box)

housing ST185-FB-POWER rev 7 BUS ZONE | contacts TBD | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **NEG** | w_gnd_id | 1/0 | Black | | POWER box gnd to ID | Ground stud ID, left kick panel (EWD ground point ID) | 1580 | ST185-EngineRoom-C | BATN |
| **NEG** | w_gnd_ig | 1/0 | Black | | POWER box gnd to IG | Ground bolt IG, right kick (EWD ground point IG) | 450 | ST185-EngineRoom-C | BATN |

### REVISIONS

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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 3 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 41 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 4 of 9)

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### GND-EA - Ground stud EA, RH front fender (EWD ground point EA)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fan2_gnd | 8 | Black | | Cond fan ground | Uprated condenser fan (GND) pin 1 | 500 | ST185-EngineRoom-C | FAN2 |
| | w_horn_lh_g | 16 | Black | | Horn LH ground | Horn LH (EWD H3) - custom DT 2-way pin 2 | 900 | ST185-EngineRoom-C | HZ |
| | w_horn_rh_g | 16 | Black | | Horn RH ground | Horn RH (EWD H4) - custom DT 2-way pin 2 | 900 | ST185-EngineRoom-C | HZ |
| | w_fan_gnd | 8 | Black | | Rad fan ground | Uprated radiator fan (GND) pin 1 | 800 | ST185-EngineRoom-C | FAN |
| | w_f8_gnd | 16 | Black | | RH front lamp gnd | RH front turn / clearance lamp F8 pin 2 pin 2 | 200 | ST185-EngineRoom-C | CLR |

### GND-EB - Ground stud EB, LH front fender (EWD ground point EB)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_f7_gnd | 16 | Black | | LH front lamp gnd | LH front turn / clearance lamp F7 pin 2 pin 2 | 200 | ST185-EngineRoom-C | CLR |
| | w_rtr_gnd_r2_t | 12 | Black | | LH retract ground | LH retract motor ground - 16 AWG pigtail to 12 AWG, at R2 | 200 | ST185-EngineRoom-C | RTM |
| | w_rtr_gnd_r3_t | 12 | Black | | RH retract ground | RH retract motor ground - 16 AWG pigtail to 12 AWG, at R3 | 1600 | ST185-EngineRoom-C | RTM |
| | w_gnd_eb | 4 | Black | | Block strap to EB | Engine block ground stud, lug 2 (Route 1 point 5) | 4050 | ST185-EngineRoom-C | BATN |

### GND-ID - Ground stud ID, left kick panel (EWD ground point ID)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_id | 1/0 | Black | | POWER box gnd to ID | POWER box bus zone (rev 7, glove box) pin NEG | 1580 | ST185-EngineRoom-C | BATN |

### GND-IG - Ground bolt IG, right kick (EWD ground point IG)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_ig | 1/0 | Black | | POWER box gnd to IG | POWER box bus zone (rev 7, glove box) pin NEG | 450 | ST185-EngineRoom-C | BATN |

### T-ENG-BLOCK-2 - Engine block ground stud, lug 2 (Route 1 point 5)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_gnd_eb | 4 | Black | | Block strap to EB | Ground stud EB, LH front fender (EWD ground point EB) | 4050 | ST185-EngineRoom-C | BATN |

### T-ENG-BLOCK-3 - Engine block ground stud, lug 3 (Route 1 point 5)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_eps_gnd | 8 | Black | | EHPS pump ground | MR-S EHPS pump plug A pin 2 | 1750 | ST185-EngineRoom-C | EPS |

### A/C compressor A4 pin 4

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_acc_clutch | 16 | Red | | LOGIC ACC 10 A fuse -> A/C clutch relay k_acc | B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B... | 1350 | ST185-EngineRoom-C | ACC |

### A/C pressure switch A5 pin 1

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_acp_1 | 18 | Yellow | | A/C press sw feed | OEM R/B No.4 10 A A/C fuse pin 2 (RH kick), V-R out pin 2 | 1600 | ST185-EngineRoom-C | ACPS |

### LH front turn / clearance lamp F7 pin 1

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_clr_f7 | 16 | Red | | LH clearance + | Clearance feed split - F7 LH + F8 RH | 900 | ST185-EngineRoom-C | CLR |

### LH front turn / clearance lamp F7 pin 2

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_f7_gnd | 16 | Black | | LH front lamp gnd | Ground stud EB, LH front fender (EWD ground point EB) | 200 | ST185-EngineRoom-C | CLR |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 4 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 42 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 5 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### LH front turn / clearance lamp F7 pin 3

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 3 | w_turn_f7 | 18 | Red | ■■■ | LH front turn + | OEM J/B No.1 1H pin 4 (left kick) pin 1H-4 | 4380 | ST185-EngineRoom-C | TURN |

### RH front turn / clearance lamp F8 pin 1

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_clr_f8 | 16 | Red | □ | RH clearance + | Clearance feed split - F7 LH + F8 RH | 900 | ST185-EngineRoom-C | CLR |

### RH front turn / clearance lamp F8 pin 2

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 2 | w_f8_gnd | 16 | Black | □ | RH front lamp gnd | Ground stud EA, RH front fender (EWD ground point EA) | 200 | ST185-EngineRoom-C | CLR |

### RH front turn / clearance lamp F8 pin 3

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 3 | w_turn_f8 | 18 | Red | ■■■ | RH front turn + | OEM J/B No.1 1H pin 8 (left kick) pin 1H-8 | 2980 | ST185-EngineRoom-C | TURN |

### Front wiper motor F9 pin 1 (cowl, driver side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_wip_hi | 14 | Red | ■■ | Wiper HIGH | OEM IE1 pin 20 (left kick, cowl side) pin 20 | 800 | ST185-EngineRoom-C | WIP |

### Front wiper motor F9 pin 2 (cowl, driver side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 2 | w_wip_lo | 14 | Red | ■■ | Wiper LOW | OEM IE1 pin 13 (left kick, cowl side) pin 13 | 800 | ST185-EngineRoom-C | WIP |

### Front wiper motor F9 pin 3 (cowl, driver side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 3 | w_wip_park | 14 | Yellow | ■■ | Wiper park sw | OEM IE1 pin 6 (left kick, cowl side) pin 6 | 800 | ST185-EngineRoom-C | WIP |

### Front wiper motor F9 pin 4 (cowl, driver side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 4 | w_wip_b | 14 | Red | ■■ | Wiper +B | OEM J/B No.1 1H pin 5 (left kick) pin 1H-5 | 800 | ST185-EngineRoom-C | WIP |

### LH headlamp H1 pin 1

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 1 | w_hl_high_h1 | 16 | Red | □■ | LH high-beam return | High-beam return join - H1 + H2 pin 1 to IE1 pin 11 | 900 | ST185-EngineRoom-C | HHI |

### LH headlamp H1 pin 2

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 2 | w_hl_low_h1 | 16 | Red | □□ | LH low-beam return | Low-beam return join - H1 + H2 pin 2 to IE1 pin 18 | 900 | ST185-EngineRoom-C | HLO |

### LH headlamp H1 pin 3

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| 3 | w_pmu_hl | 14 | Red | ■■■ | LOGIC HL 15 A fuse -> HEAD LH relay k_hl | B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 87 | 3000 | ST185-EngineRoom-C | HL |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 5 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 43 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 6 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### RH headlamp H2 pin 1

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_hl_high_h2 | 16 | Red | □■ | RH high-beam return | High-beam return join - H1 + H2 pin 1 to IE1 pin 11 | 900 | ST185-EngineRoom-C | HHI |

### RH headlamp H2 pin 2

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_hl_low_h2 | 16 | Red | □□ | RH low-beam return | Low-beam return join - H1 + H2 pin 2 to IE1 pin 18 | 900 | ST185-EngineRoom-C | HLO |

### RH headlamp H2 pin 3

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | w_pmu_hr | 14 | Red | ■□ | LOGIC HR 15 A fuse -> HEAD RH relay k_hr | B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 87 | 1600 | ST185-EngineRoom-C | HR |

### OEM IE1 pin 10 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **10** | w_pdb_am1 | 8 | Red | ■□ | AMI AM1 40 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 1580 | ST185-EngineRoom-C | AM1 |

### OEM IE1 pin 11 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **11** | w_hl_high | 14 | Red | □■ | High-beam return | High-beam return join - H1 + H2 pin 1 to IE1 pin 11 | 3680 | ST185-EngineRoom-C | HHI |

### OEM IE1 pin 13 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **13** | w_wip_lo | 14 | Red | ■□ | Wiper LOW | Front wiper motor F9 pin 2 (cowl, driver side) pin 2 | 800 | ST185-EngineRoom-C | WIP |

### OEM IE1 pin 14 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **14** | w_rtr_r2_4 | 12 | Red | ■ | LH retract motor 4 | LH retract motor feed - 12 AWG to 16 AWG pigtail, at R2 | 4380 | ST185-EngineRoom-C | RTM |

### OEM IE1 pin 15 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **15** | w_rtr_r2_2 | 16 | Red | ■ | LH retract motor 2 | LH retract motor R2 pin 2 pin 2 | 4380 | ST185-EngineRoom-C | RTM |

### OEM IE1 pin 16 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **16** | w_rtr_open | 18 | Yellow | ■ | Retract open limit | Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16 | 3680 | ST185-EngineRoom-C | RTM |

### OEM IE1 pin 17 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **17** | w_pdb_am2 | 10 | Red | ■□■ | AMI AM2 30 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 1580 | ST185-EngineRoom-C | AM2 |

### OEM IE1 pin 18 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **18** | w_hl_low | 14 | Red | □□ | Low-beam return | Low-beam return join - H1 + H2 pin 2 to IE1 pin 18 | 3680 | ST185-EngineRoom-C | HLO |

### REVISIONS

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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 6 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 44 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 7 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### OEM IE1 pin 20 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **20** | w_wip_hi | 14 | Red | | Wiper HIGH | Front wiper motor F9 pin 1 (cowl, driver side) pin 1 | 800 | ST185-EngineRoom-C | WIP |

### OEM IE1 pin 3 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | w_pmu_rtr | 10 | Red | | AMI RTR 30 A fuse | B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse... | 1580 | ST185-EngineRoom-C | RTR |

### OEM IE1 pin 6 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **6** | w_wip_park | 14 | Yellow | | Wiper park sw | Front wiper motor F9 pin 3 (cowl, driver side) pin 3 | 800 | ST185-EngineRoom-C | WIP |

### OEM IE1 pin 7 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **7** | w_rtr_r3_4 | 12 | Red | | RH retract motor 4 | RH retract motor feed - 12 AWG to 16 AWG pigtail, at R3 | 2980 | ST185-EngineRoom-C | RTM |

### OEM IE1 pin 8 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **8** | w_rtr_r3_2 | 16 | Red | | RH retract motor 2 | RH retract motor R3 pin 2 pin 2 | 2980 | ST185-EngineRoom-C | RTM |

### OEM IE1 pin 9 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **9** | w_rtr_close | 18 | Yellow | | Retract close limit | Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9 | 3680 | ST185-EngineRoom-C | RTM |

### OEM IG2 pin 16 (under the ECU, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **16** | w_alt_l_c | 20 | White | | Alt L charge lamp | Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout) pin 2 | 3140 | ST185-EngineRoom-C | ALTL |

### OEM J/B No.1 1H pin 4 (left kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1H-4** | w_turn_f7 | 18 | Red | | LH front turn + | LH front turn / clearance lamp F7 pin 3 pin 3 | 4380 | ST185-EngineRoom-C | TURN |

### OEM J/B No.1 1H pin 5 (left kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1H-5** | w_wip_b | 14 | Red | | Wiper +B | Front wiper motor F9 pin 4 (cowl, driver side) pin 4 | 800 | ST185-EngineRoom-C | WIP |

### OEM J/B No.1 1H pin 6 (left kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1H-6** | w_clr | 16 | Red | | Clearance lamp feed | Clearance feed split - F7 LH + F8 RH | 3680 | ST185-EngineRoom-C | CLR |

### OEM J/B No.1 1H pin 8 (left kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1H-8** | w_turn_f8 | 18 | Red | | RH front turn + | RH front turn / clearance lamp F8 pin 3 pin 3 | 2980 | ST185-EngineRoom-C | TURN |

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| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 45 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 8 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### OEM J/B No.1 1H pin 2 (left kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_pmu_dome | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC DM 20 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 1580 | ST185-EngineRoom-C | DM |

### LH retract motor R2 pin 1

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_rtr_open_r2 | 18 | Yellow | | Retract open limit | Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16 | 900 | ST185-EngineRoom-C | RTM |

### LH retract motor R2 pin 2

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_rtr_r2_2 | 16 | Red | | LH retract motor 2 | OEM IE1 pin 15 (left kick, cowl side) pin 15 | 4380 | ST185-EngineRoom-C | RTM |

### LH retract motor R2 pin 3

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | w_rtr_close_r2 | 18 | Yellow | | Retract close limit | Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9 | 900 | ST185-EngineRoom-C | RTM |

### LH retract motor R2 pin 4

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_rtr_r2_4_p | 16 | Red | | LH retract motor 4 | LH retract motor feed - 12 AWG to 16 AWG pigtail, at R2 | 200 | ST185-EngineRoom-C | RTM |

### LH retract motor R2 pin 5

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | w_rtr_gnd_r2 | 16 | Black | | LH retract ground | LH retract motor ground - 16 AWG pigtail to 12 AWG, at R2 | 200 | ST185-EngineRoom-C | RTM |

### RH retract motor R3 pin 1

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_rtr_open_r3 | 18 | Yellow | | Retract open limit | Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16 | 900 | ST185-EngineRoom-C | RTM |

### RH retract motor R3 pin 2

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_rtr_r3_2 | 16 | Red | | RH retract motor 2 | OEM IE1 pin 8 (left kick, cowl side) pin 8 | 2980 | ST185-EngineRoom-C | RTM |

### RH retract motor R3 pin 3

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | w_rtr_close_r3 | 18 | Yellow | | Retract close limit | Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9 | 900 | ST185-EngineRoom-C | RTM |

### RH retract motor R3 pin 4

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **4** | w_rtr_r3_4_p | 16 | Red | | RH retract motor 4 | RH retract motor feed - 12 AWG to 16 AWG pigtail, at R3 | 200 | ST185-EngineRoom-C | RTM |

### RH retract motor R3 pin 5

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **5** | w_rtr_gnd_r3 | 16 | Black | | RH retract ground | RH retract motor ground - 16 AWG pigtail to 12 AWG, at R3 | 200 | ST185-EngineRoom-C | RTM |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 8 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 46 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 9 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### OEM W-L lead at old R/B2 pin 2 (left kick)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_rb2_wl | 14 | Red | | AMI BUS1 50 A fuse -> LOGIC WL 15 A fuse | B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ... | 1580 | ST185-EngineRoom-C | WL |

### OEM R/B No.4 10 A A/C fuse pin 2 (RH kick), V-R out

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_acp_1 | 18 | Yellow | | A/C press sw feed | A/C pressure switch A5 pin 1 pin 1 | 1600 | ST185-EngineRoom-C | ACPS |

### OEM washer W1 pin 1 cowl lead (firewall, cabin side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_wash_neg | 14 | Red | | Washer switch side | Washer motor W1 pin 1 (RH front corner, low) pin 1 | 1300 | ST185-EngineRoom-C | WSH |

### Washer motor W1 pin 1 (RH front corner, low)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_wash_neg | 14 | Red | | Washer switch side | OEM washer W1 pin 1 cowl lead (firewall, cabin side) pin 1 | 1300 | ST185-EngineRoom-C | WSH |

### OEM washer W1 pin 2 cowl lead (firewall, cabin side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_wash_pos | 14 | Red | | Washer motor +12 V | Washer motor W1 pin 2 (RH front corner, low) pin 2 | 1300 | ST185-EngineRoom-C | WSH |

### Washer motor W1 pin 2 (RH front corner, low)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **2** | w_wash_pos | 14 | Red | | Washer motor +12 V | OEM washer W1 pin 2 cowl lead (firewall, cabin side) pin 2 | 1300 | ST185-EngineRoom-C | WSH |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS  (part 9 ...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 47 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK  (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### ALT-PLUG - Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout)

housing 90980-10341 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **3** | w_ig_ks_out | 16 | Red | | Alt IG excite | Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin ALT2 | 5110 | ST185-HeavyDC | ALTIG |

### BATT-RELAY-BLOCK - Battery relay block - fuel pump relay + its fuse, trunk, passenger side between the rear wall and the battery (cad/fusebox rev 5, DECISIONS R75, Z4)

housing ST185-FB-BATT rev 5 | contacts TBD | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **BATT** | w_fp_feed | 10 | White | | Battery relay box + | Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin MAIN2 | 200 | ST185-HeavyDC | FP |
| **FP** | w_fp_30 | 10 | Red | | FP 40 A fuse (battery relay block) | B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ... | 200 | ST185-HeavyDC | FP |

### JUMP-POST - Engine-bay jump post

housing TBD | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_jump | 2 | Red | | Jump post feed | Starter B+ stud, lug 3 (Route 1 point 2) | 410 | ST185-HeavyDC | BAT |

### K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts TBD | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **30** | w_fp_30 | 10 | Red | | FP 40 A fuse (battery relay block) | Battery relay block - fuel pump relay + its fuse, trunk, passenger side between the rear wall an... | 200 | ST185-HeavyDC | FP |
| **87a** | | | | | spare, not used | | | | |

### KILL-SW - Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1)

housing 52-45780 | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **ALT1** | w_ig_ks_in | 16 | Red | | Alt IG to kill sw | OEM IG2 pin 15 (under the ECU, cowl side) pin 15 | 3130 | ST185-HeavyDC | ALTIG |
| **ALT2** | w_ig_ks_out | 16 | Red | | Alt IG excite | Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout) pin 3 | 5110 | ST185-HeavyDC | ALTIG |
| **MAIN1** | w_trunk_pos | 1/0 | Red | | battery + | Battery positive terminal post, trunk RH rear (Route 1 point 1) | 200 | ST185-HeavyDC | BAT |
| **MAIN2** | w_fp_feed | 10 | White | | Battery relay box + | Battery relay block - fuel pump relay + its fuse, trunk, passenger side between the rear wall an... | 200 | ST185-HeavyDC | FP |
| **MAIN2** | w_ks_h1 | 1/0 | Red | | battery + | POWER box bus zone (rev 7, glove box) pin BAT | 2650 | ST185-HeavyDC | BAT |

### POWER-STUDS - POWER box bus zone (rev 7, glove box)

housing ST185-FB-POWER rev 7 BUS ZONE | contacts TBD | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **ALT** | w_h3_radlok_pos | 1/0 | Red | | Starter feed (H3) | RADLOK + firewall pin 1 | 600 | ST185-HeavyDC | BAT |
| **BAT** | w_ks_h1 | 1/0 | Red | | battery + | Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin MAIN2 | 2650 | ST185-HeavyDC | BAT |
| **NEG** | w_h6a_radlok_neg | 1/0 | Black | | Engine ground (H6a) | RADLOK - firewall pin 1 | 600 | ST185-HeavyDC | BATN |
| **NEG** | w_trunk_neg | 1/0 | Black | | Battery - main | Battery negative terminal post, trunk RH rear (Route 1 point 1) | 2650 | ST185-HeavyDC | BATN |

### RADLOK-NEG-CAB - RADLOK - firewall

housing RL00801-50BK | contacts INTEGRAL | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_h6a_radlok_neg | 1/0 | Black | | Engine ground (H6a) | POWER box bus zone (rev 7, glove box) pin NEG | 600 | ST185-HeavyDC | BATN |
| **1** | w_rln | 1/0 | Black | | Engine ground (H6a) | RADLOK - engine pin 1 | 2760 | ST185-HeavyDC | BATN |

### RADLOK-NEG-ENG - RADLOK - engine

housing RL00801-50BK | contacts INTEGRAL | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_rln_blk | 1/0 | Black | | Engine ground (H6a) | Engine block ground stud, lug 1 (Route 1 point 5) | 890 | ST185-HeavyDC | BATN |
| **1** | w_rln | 1/0 | Black | | Engine ground (H6a) | RADLOK - firewall pin 1 | 2760 | ST185-HeavyDC | BATN |

### RADLOK-POS-CAB - RADLOK + firewall

housing RL00801-50RE | contacts INTEGRAL | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_h3_radlok_pos | 1/0 | Red | | Starter feed (H3) | POWER box bus zone (rev 7, glove box) pin ALT | 600 | ST185-HeavyDC | BAT |
| **1** | w_rl_pos | 1/0 | Red | | Starter feed (H3) | RADLOK + engine pin 1 | 2760 | ST185-HeavyDC | BAT |

### REVISIONS

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|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK  (part 1 o...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 48 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK  (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### RADLOK-POS-ENG - RADLOK + engine

housing RL00801-50RE | contacts INTEGRAL | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_rlp_str | 1/0 | Red | | Starter feed (H3) | Starter B+ stud, lug 1 (Route 1 point 2) | 1700 | ST185-HeavyDC | BAT |
| **1** | w_rl_pos | 1/0 | Red | | Starter feed (H3) | RADLOK + firewall pin 1 | 2760 | ST185-HeavyDC | BAT |

### BATT-POS - Battery positive terminal post, trunk RH rear (Route 1 point 1)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_trunk_pos | 1/0 | Red | | battery + | Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin MAIN1 | 200 | ST185-HeavyDC | BAT |

### T-ALT-B - Alternator B+ stud, rear RH of engine (Route 1 point 3)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_alt | 2 | Red | | Alternator B+ | Starter B+ stud, lug 2 (Route 1 point 2) | 2000 | ST185-HeavyDC | BAT |

### T-ENG-BLOCK - Engine block ground stud, lug 1 (Route 1 point 5)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rln_blk | 1/0 | Black | | Engine ground (H6a) | RADLOK - engine pin 1 | 890 | ST185-HeavyDC | BATN |

### T-STARTER-B - Starter B+ stud, lug 1 (Route 1 point 2)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_rlp_str | 1/0 | Red | | Starter feed (H3) | RADLOK + engine pin 1 | 1700 | ST185-HeavyDC | BAT |

### T-STARTER-B-2 - Starter B+ stud, lug 2 (Route 1 point 2)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_alt | 2 | Red | | Alternator B+ | Alternator B+ stud, rear RH of engine (Route 1 point 3) | 2000 | ST185-HeavyDC | BAT |

### T-STARTER-B-3 - Starter B+ stud, lug 3 (Route 1 point 2)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_jump | 2 | Red | | Jump post feed | Engine-bay jump post pin 1 | 410 | ST185-HeavyDC | BAT |

### T-TRUNK-NEG - Battery negative terminal post, trunk RH rear (Route 1 point 1)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_trunk_neg | 1/0 | Black | | Battery - main | POWER box bus zone (rev 7, glove box) pin NEG | 2650 | ST185-HeavyDC | BATN |

### OEM IG2 pin 15 (under the ECU, cowl side)

factory lead tap (EWD locator in the signal)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **15** | w_ig_ks_in | 16 | Red | | Alt IG to kill sw | Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin ALT1 | 3130 | ST185-HeavyDC | ALTIG |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

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HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK  (part 2 o...

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 49 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

## REAR FUEL - FUEL PUMP AND LEVEL SENDER

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### FUEL-LEVEL - Fuel Level Sender - DT 2-way (Dan rev 2, 2026-10-10 - no OEM connector)

housing DT04-2P + W2P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_fl_sig_r | 20 | White | | Fuel level signal | IX-FUEL inline to fuel sender (trunk LH) pin 1 | 500 | ST185-RearFuel | FLVL |
| **2** | w45_r | 20 | Blue | | Fuel level ground | IX-FUEL inline to fuel sender (trunk LH) pin 2 | 500 | ST185-RearFuel | FLVL |

### FUEL-PUMP - Fuel Pump

housing TBD | contacts TBD | pins per TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_fpump_r | 10 | Red | | Fuel pump +12 V | B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ... | 1500 | ST185-RearFuel | FP |
| **2** | w_fpg | 12 | Black | | Fuel pump ground | Ground stud, trunk LH side (EWD ID TBD, M6 assumed) | 200 | ST185-RearFuel | FP |

### IX-FUEL-S - IX-FUEL inline to fuel sender (trunk LH)

housing DT06-2S + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | w_fl_sig_r | 20 | White | | Fuel level signal | Fuel Level Sender - DT 2-way (Dan rev 2, 2026-10-10 - no OEM connector) pin 1 | 500 | ST185-RearFuel | FLVL |
| **2** | w45_r | 20 | Blue | | Fuel level ground | Fuel Level Sender - DT 2-way (Dan rev 2, 2026-10-10 - no OEM connector) pin 2 | 500 | ST185-RearFuel | FLVL |

### K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts TBD | pins per ISO 7588 relay terminal numbers

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **87** | w_fpump_r | 10 | Red | | Fuel pump +12 V | Fuel Pump pin 1 | 1500 | ST185-RearFuel | FP |

### T-TRUNK-FP-GND - Ground stud, trunk LH side (EWD ID TBD, M6 assumed)

terminal TBD

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | w_fpg | 12 | Black | | Fuel pump ground | Fuel Pump pin 2 | 200 | ST185-RearFuel | FP |

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
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| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

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REAR FUEL - FUEL PUMP AND LEVEL SENDER

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 50 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## FRONT WHEEL SPEED - FL / FR SENSORS TO THE FRONT VRC

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-WS-F-SHIELD - Sensor screen join - FL + FR screens to the VRC IN pin 3 drain

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_fl_sh | | Shield | | FL sensor shield | cab_fl screen (floats at the device end) | 200 | ST185-WheelSpeed-Front | WSS_FL |
| | cab_fr_sh | | Shield | | FR sensor shield | cab_fr screen (floats at the device end) | 200 | ST185-WheelSpeed-Front | WSS_FR |
| | w_ws_f_shield | 20 | Black | | Front sensor shields | VRC front IN plug pin 3 | 200 | ST185-WheelSpeed-Front | VRC_F_DRAIN |

### IX-WS-F-P - IX-WS-F

housing DT04-6P + W6P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_fout_5v | 20 | Red | | VRC-F +5V | VRC front OUT plug pin 1 | 200 | ST185-WheelSpeed-Front | VRC_F_5V |
| **2** | cab_fout_gnd | 20 | Yellow | | VRC-F GND | VRC front OUT plug pin 2 | 200 | ST185-WheelSpeed-Front | VRC_F_GND |
| **3** | cab_fout_l | 20 | Green | | FL conditioned out | VRC front OUT plug pin 4 | 200 | ST185-WheelSpeed-Front | WSS_FL |
| **4** | cab_fout_r | 20 | Blue | | FR conditioned out | VRC front OUT plug pin 5 | 200 | ST185-WheelSpeed-Front | WSS_FR |
| **5** | cab_fout_sh | | Shield | | WS-F screen drain | VRC front OUT plug pin 3 | 200 | ST185-WheelSpeed-Front | VRC_F_DRAIN |
| **6** | | | | | spare, sealing plug | | | | |

### VRC-F-IN - VRC front IN plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_fl_p | 20 | Blue | | FL wheel speed + | ABS Wheel Speed FL pin 1 | 1670 | ST185-WheelSpeed-Front | WSS_FL |
| **2** | cab_fl_n | 20 | White | | FL wheel speed - | ABS Wheel Speed FL pin 2 | 1670 | ST185-WheelSpeed-Front | WSS_FL |
| **3** | w_ws_f_shield | 20 | Black | | Front sensor shields | Sensor screen join - FL + FR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Front | VRC_F_DRAIN |
| **4** | cab_fr_p | 20 | Blue | | FR wheel speed + | ABS Wheel Speed FR pin 1 | 1560 | ST185-WheelSpeed-Front | WSS_FR |
| **5** | cab_fr_n | 20 | White | | FR wheel speed - | ABS Wheel Speed FR pin 2 | 1560 | ST185-WheelSpeed-Front | WSS_FR |

### VRC-F-OUT - VRC front OUT plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_fout_5v | 20 | Red | | VRC-F +5V | IX-WS-F pin 1 | 200 | ST185-WheelSpeed-Front | VRC_F_5V |
| **2** | cab_fout_gnd | 20 | Yellow | | VRC-F GND | IX-WS-F pin 2 | 200 | ST185-WheelSpeed-Front | VRC_F_GND |
| **3** | cab_fout_sh | | Shield | | WS-F screen drain | IX-WS-F pin 5 | 200 | ST185-WheelSpeed-Front | VRC_F_DRAIN |
| **4** | cab_fout_l | 20 | Green | | FL conditioned out | IX-WS-F pin 3 | 200 | ST185-WheelSpeed-Front | WSS_FL |
| **5** | cab_fout_r | 20 | Blue | | FR conditioned out | IX-WS-F pin 4 | 200 | ST185-WheelSpeed-Front | WSS_FR |

### WSS-FL - ABS Wheel Speed FL

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_fl_p | 20 | Blue | | FL wheel speed + | VRC front IN plug pin 1 | 1670 | ST185-WheelSpeed-Front | WSS_FL |
| **2** | cab_fl_n | 20 | White | | FL wheel speed - | VRC front IN plug pin 2 | 1670 | ST185-WheelSpeed-Front | WSS_FL |

### WSS-FR - ABS Wheel Speed FR

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_fr_p | 20 | Blue | | FR wheel speed + | VRC front IN plug pin 4 | 1560 | ST185-WheelSpeed-Front | WSS_FR |
| **2** | cab_fr_n | 20 | White | | FR wheel speed - | VRC front IN plug pin 5 | 1560 | ST185-WheelSpeed-Front | WSS_FR |

### CAB-FL cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_fl_sh | | Shield | | FL sensor shield | Sensor screen join - FL + FR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Front | WSS_FL |

### CAB-FR cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_fr_sh | | Shield | | FR sensor shield | Sensor screen join - FL + FR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Front | WSS_FR |

### REVISIONS

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| I | 2026-10-27 | DG | Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra... |
| J | 2026-10-28 | DG | ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO... |
| K | 2026-10-29 | DG | Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ... |
| L | 2026-10-29 | DG | Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn... |

**ST185-harness - every wiring harness of the car, one drawing set**

FRONT WHEEL SPEED - FL / FR SENSORS TO THE FRONT VRC

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-PIN | L | 51 OF 52 | 432 x 279 mm | DG | 2026-10-29 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

## REAR WHEEL SPEED - RL / RR SENSORS TO THE REAR VRC

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

### SP-WS-R-SHIELD - Sensor screen join - RL + RR screens to the VRC IN pin 3 drain

splice: solder splice

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_rl_sh | | Shield | | RL sensor shield | cab_rl screen (floats at the device end) | 200 | ST185-WheelSpeed-Rear | WSS_RL |
| | cab_rr_sh | | Shield | | RR sensor shield | cab_rr screen (floats at the device end) | 200 | ST185-WheelSpeed-Rear | WSS_RR |
| | w_ws_r_shield | 20 | Black | | Rear sensor shields | VRC rear IN plug pin 3 | 200 | ST185-WheelSpeed-Rear | VRC_R_DRAIN |

### IX-WS-R-P - IX-WS-R

housing DT04-6P + W6P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_rout_5v | 20 | Red | | VRC-R +5V | VRC rear OUT plug pin 1 | 200 | ST185-WheelSpeed-Rear | VRC_R_5V |
| **2** | cab_rout_gnd | 20 | Yellow | | VRC-R GND | VRC rear OUT plug pin 2 | 200 | ST185-WheelSpeed-Rear | VRC_R_GND |
| **3** | cab_rout_l | 20 | Green | | Rear left wheel spd | VRC rear OUT plug pin 4 | 200 | ST185-WheelSpeed-Rear | WSS_RL |
| **4** | cab_rout_r | 20 | Blue | | Rear right wheel spd | VRC rear OUT plug pin 5 | 200 | ST185-WheelSpeed-Rear | WSS_RR |
| **5** | cab_rout_sh | | Shield | | WS-R screen drain | VRC rear OUT plug pin 3 | 200 | ST185-WheelSpeed-Rear | VRC_R_DRAIN |
| **6** | | | | | spare, sealing plug | | | | |

### VRC-R-IN - VRC rear IN plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_rl_p | 20 | Blue | | RL wheel speed + | ABS Wheel Speed RL pin 1 | 3570 | ST185-WheelSpeed-Rear | WSS_RL |
| **2** | cab_rl_n | 20 | White | | RL wheel speed - | ABS Wheel Speed RL pin 2 | 3570 | ST185-WheelSpeed-Rear | WSS_RL |
| **3** | w_ws_r_shield | 20 | Black | | Rear sensor shields | Sensor screen join - RL + RR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Rear | VRC_R_DRAIN |
| **4** | cab_rr_p | 20 | Blue | | RR wheel speed + | ABS Wheel Speed RR pin 1 | 4470 | ST185-WheelSpeed-Rear | WSS_RR |
| **5** | cab_rr_n | 20 | White | | RR wheel speed - | ABS Wheel Speed RR pin 2 | 4470 | ST185-WheelSpeed-Rear | WSS_RR |

### VRC-R-OUT - VRC rear OUT plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_rout_5v | 20 | Red | | VRC-R +5V | IX-WS-R pin 1 | 200 | ST185-WheelSpeed-Rear | VRC_R_5V |
| **2** | cab_rout_gnd | 20 | Yellow | | VRC-R GND | IX-WS-R pin 2 | 200 | ST185-WheelSpeed-Rear | VRC_R_GND |
| **3** | cab_rout_sh | | Shield | | WS-R screen drain | IX-WS-R pin 5 | 200 | ST185-WheelSpeed-Rear | VRC_R_DRAIN |
| **4** | cab_rout_l | 20 | Green | | Rear left wheel spd | IX-WS-R pin 3 | 200 | ST185-WheelSpeed-Rear | WSS_RL |
| **5** | cab_rout_r | 20 | Blue | | Rear right wheel spd | IX-WS-R pin 4 | 200 | ST185-WheelSpeed-Rear | WSS_RR |

### WSS-RL - ABS Wheel Speed RL

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_rl_p | 20 | Blue | | RL wheel speed + | VRC rear IN plug pin 1 | 3570 | ST185-WheelSpeed-Rear | WSS_RL |
| **2** | cab_rl_n | 20 | White | | RL wheel speed - | VRC rear IN plug pin 2 | 3570 | ST185-WheelSpeed-Rear | WSS_RL |

### WSS-RR - ABS Wheel Speed RR

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| **1** | cab_rr_p | 20 | Blue | | RR wheel speed + | VRC rear IN plug pin 4 | 4470 | ST185-WheelSpeed-Rear | WSS_RR |
| **2** | cab_rr_n | 20 | White | | RR wheel speed - | VRC rear IN plug pin 5 | 4470 | ST185-WheelSpeed-Rear | WSS_RR |

### CAB-RL cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_rl_sh | | Shield | | RL sensor shield | Sensor screen join - RL + RR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Rear | WSS_RL |

### CAB-RR cable screen (floats at the device end)

cable screen

| PIN | WIRE | AWG | COLOUR | BANDS | SIGNAL | OTHER END | LEN MM | BUILT WITH | CIRCUIT |
|---|---|---|---|---|---|---|---|---|---|
| | cab_rr_sh | | Shield | | RR sensor shield | Sensor screen join - RL + RR screens to the VRC IN pin 3 drain | 200 | ST185-WheelSpeed-Rear | WSS_RR |

### REVISIONS

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